

A THESIS REPORT
On
Fetal Down Syndrome Risk Prediction via Multimodal Fusion with
Feature-Based Modeling

Submitted by

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Under the Guidance of

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Declaration of Student

I hereby declare that the project entitled “Fetal Down Syndrome Risk Prediction via Multimodal Fusion with Feature-Based Modeling” submitted for the Thesis Report II CSE620P, is my original work and the project has not formed the basis for the award of any other degree, diploma, fellowship or any other similar titles.

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This is to certify that the project titled “Fetal Down Syndrome Risk Prediction via Multimodal Fusion with Feature-Based Modeling” is the bonafide work carried out by Joyeeta Das students of M.Tech Department of Computer Science and Engineering, The ICFAI University, Tripura during the Thesis report 2026, in partial fulfillment of the requirements for the award of the degree of and that the project has not formed the basis for the award previously of any other degree, diploma, fellowship or any other similar title.

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ABSTRACT

Down syndrome arises due to the presence of an additional copy of 21st chromosome. It causes delayed anatomical development and congenital abnormalities in fetus. Timely assessment of risk is crucial for clinical evaluation and decision making. The usual way is Conventional ultrasound-based screening, which relies heavily on manual interpretation of biomarkers and can be time-consuming. The study aims to develop an automated and robust framework for Down syndrome risk assessment by integrating multimodal fetal imaging. The system integrates three modalities: ultrasound, MRI, and synthetically generated MRI through cross-modality learning. Ultrasound images are processed using No New U-Net (nnU-Net) for segmentation, followed by biomarker extraction such as nuchal translucency, nasal structure, brain region, and posterior fossa. Synthetic MRI is generated using the Contrastive Unpaired Translation(CUT) model to enhance structural representation. In addition, a lightweight and interpretable feature-based model is introduced to support transparent risk prediction. Experimental findings indicate that the framework achieves consistent and accurate risk prediction, with synthetic MRI improving performance over ultrasound alone. The proposed approach demonstrates the potential of combining multimodal imaging with interpretable modeling for practical prenatal screening applications.

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CHAPTER 1

INTRODUCTION

Prenatal screening is an essential component of modern maternal care and plays a crucial role in monitoring fetal development during pregnancy. Proactive identification of fetal abnormalities allows clinicians to provide appropriate medical guidance, recommend further diagnostic procedures, and improve prenatal management. Among the various chromosomal disorders observed during prenatal screening, Down syndrome (Trisomy 21) is one of the most common genetic conditions. It occurs due to the presence of three copies of chromosome 21 instead of two and is associated with cognitive impairment, developmental delays, and a variety of structural abnormalities.[1]

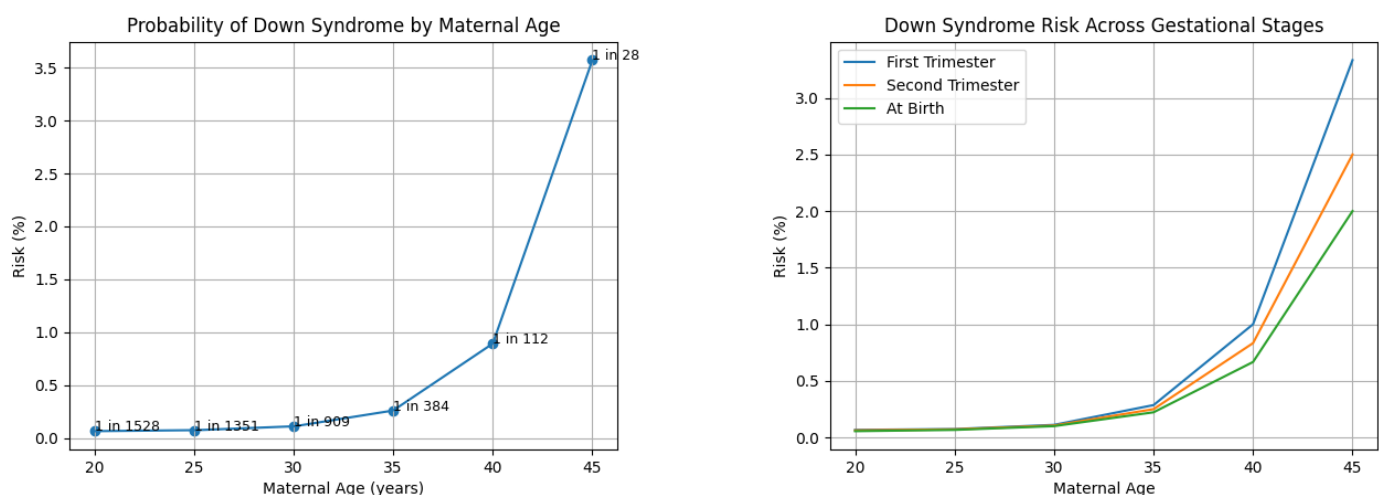


Figure 1: Probability of DS By maternal age and Risk across Gestational stages

Fig 1. shows the Probability of DS By maternal age and Risk across Gestational stages. During the first trimester of pregnancy, particularly between 11 and 14 weeks of gestation, ultrasound imaging is widely used to examine fetal anatomical structures.[2] Certain anatomical markers visible in ultrasound images have been linked to a higher risk of Down syndrome. One of the most widely studied indicators is Nuchal Translucency (NT) is the small fluid space behind the fetal neck.[3] NT thickness has been linked to chromosomal abnormalities, including Down syndrome. In addition to NT measurements, other anatomical features such as nasal bone visibility, brain region characteristics, and posterior fossa structures provide important information about fetal development and potential abnormalities.[1,3]

Traditionally, these biomarkers are measured manually by trained sonographers through visual inspection of ultrasound images. Although this approach is commonly used in clinical practice, manual measurement has several limitations.[4,5] The process is time taken and highly dependent on the experience of the clinician. Ultrasound images often contain noise, artifacts, and variations in image

quality, which can affect the accuracy of measurements. Furthermore, differences in interpretation between observers may lead to inconsistent screening results.[5]

In the last few years, medical imaging has started to incorporate artificial intelligence (AI) and machine learning methods and It has a significant potential in medical image analysis. Deep learning methods are often used for medical image segmentation tasks, feature extraction, and disease classification. Automated image segmentation techniques can accurately identify anatomical structures within medical images, enabling the extraction of quantitative biomarkers that can be used for clinical analysis and predictive modeling.

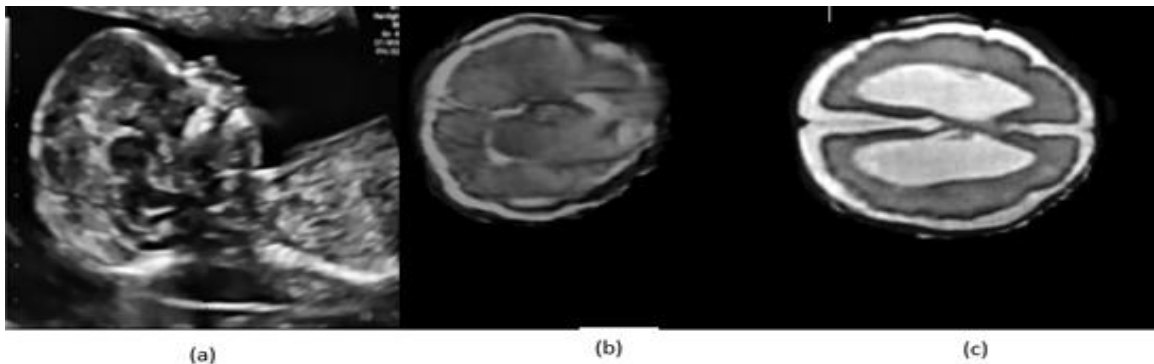


Figure 1.2 : Multimodal fetal imaging: (a) Ultrasound, (b) MRI, and (c) Synthetic MRI generated using cross-modality learning.

Another imaging modality that provides detailed anatomical information is Magnetic Resonance Imaging (MRI). Compared to ultrasound imaging, MRI offers improved soft tissue contrast and allows better visualization of fetal brain structures. However, MRI is less commonly used for early prenatal screening due to higher cost, longer acquisition time, and limited availability. In addition, publicly available datasets containing both ultrasound and MRI images for the same fetal cases are limited, which presents challenges for multimodal analysis.

To address these limitations, recent research has explored cross-modality domain adaptation techniques, where deep learning models are used to translate images from one imaging modality to another. Such approaches enable the generation of synthetic images that resemble another modality, allowing researchers to utilize complementary information even when paired multimodal datasets are unavailable.[6] Motivated by these developments, this thesis proposes an AI-based framework for Down syndrome risk prediction using ultrasound imaging, MRI analysis, and synthetic MRI generation through domain adaptation. The proposed system first performs automated segmentation of fetal ultrasound images using the nnUNet deep learning architecture, which identifies relevant anatomical regions. From the segmentation masks, important biomarkers such as NT area, nasal structure area,

brain region area, and posterior fossa measurements are extracted. The extracted features are then fed into machine learning models that estimate Down syndrome risk levels.[6,7]

In addition to ultrasound analysis, fetal MRI datasets are incorporated into the study for multimodal feature analysis. MRI volumes are preprocessed and converted into two-dimensional slices for further processing. To overcome the limitation of limited multimodal datasets, a contrastive unpaired image translation model (CUT) is used to generate synthetic MRI images from ultrasound images, enabling cross-modality analysis.

By integrating ultrasound imaging, MRI data, and synthetic MRI generation the introduced framework is designed to provide an automated and data-driven approach for Down syndrome risk prediction. Such AI-assisted systems have the potential to support clinicians in prenatal screening by providing objective and reproducible analysis of fetal anatomical biomarkers.[18]

1.1 Problem Definition

Early detection of chromosomal abnormalities during pregnancy is a critical component of prenatal healthcare. Among various genetic conditions, Down syndrome (Trisomy 21) is one of the most frequently observed chromosomal disorders worldwide. Early screening for Down syndrome enables clinicians to recommend appropriate diagnostic tests and provide informed guidance to expectant parents. In clinical practice, the first trimester ultrasound examination between 11 and 14 weeks of gestation plays a significant role in identifying anatomical markers associated with Down syndrome risk.

Several fetal anatomical biomarkers observable in ultrasound images are commonly used for this purpose. Among them, Nuchal Translucency (NT) thickness, nasal bone visibility, brain region characteristics, and posterior fossa structures are widely recognized indicators associated with chromosomal abnormalities. Traditionally, these biomarkers are measured manually by experienced sonographers using ultrasound imaging systems. However, manual measurement presents several limitations. The process is time taken and relies heavily on the expertise of the clinician, which may lead to variability in measurements and inconsistent interpretation of results. In addition, ultrasound images often contain noise, artifacts, and varying levels of contrast that can make precise measurement challenging.

Another limitation in prenatal research is the lack of publicly available datasets containing confirmed Down syndrome labels along with detailed fetal imaging data. This restricts the development of fully supervised diagnostic systems for automated Down syndrome detection. As a result, many research studies rely on biomarker-based risk estimation approaches, where anatomical measurements

extracted from fetal images are used to estimate relative risk levels rather than performing direct diagnosis.

Furthermore, while Magnetic Resonance Imaging (MRI) provides improved visualization of fetal brain structures, it is not routinely used in early prenatal screening due to higher cost, longer acquisition time, and limited accessibility. Additionally, datasets containing both ultrasound and MRI images for the same fetal cases are scarce, which makes multimodal analysis challenging.

Therefore, there is a need to develop an automated computational framework capable of analyzing fetal medical images, extracting clinically relevant biomarkers, and estimating Down syndrome risk levels using artificial intelligence techniques. Such a system can assist clinicians by providing objective and reproducible assessments based on imaging data.

In this thesis, the problem addressed is the development of an AI-based Down syndrome risk prediction framework that integrates automated ultrasound image segmentation, biomarker extraction, machine learning-based risk classification, and cross-modality domain adaptation for synthetic MRI generation. The goal of the proposed system is to support prenatal screening by enabling automated analysis of fetal anatomical structures and estimating potential Down syndrome risk levels.

1.2 Project Overview / Specifications

This project aims to develop a system based on artificial intelligence for Down syndrome risk prediction using fetal medical imaging data. The proposed framework integrates ultrasound imaging, magnetic resonance imaging (MRI), and synthetic MRI generation through cross-modality domain adaptation in order to build a comprehensive computational pipeline for automated prenatal risk assessment.

The proposed system is designed to automate several key stages involved in fetal image analysis, including anatomical segmentation, biomarker extraction, machine learning-based risk classification, and multimodal image analysis. By combining these components, the framework aims to provide a reliable and reproducible approach for estimating Down syndrome risk levels from fetal imaging data. The workflow of the proposed system begins with the analysis of ultrasound (USG) images, which are commonly used in first trimester prenatal screening. In the first stage, ultrasound images are processed using the nnUNet deep learning segmentation framework, a widely used architecture for biomedical image segmentation. The segmentation model automatically identifies important fetal anatomical structures within the ultrasound images. These segmented regions allow the system to locate clinically relevant anatomical areas that are associated with Down syndrome risk.

Once the segmentation process is completed, the system performs biomarker extraction from the segmentation masks. Quantitative anatomical measurements are computed for several fetal structures, including Nuchal Translucency (NT) area, nasal region area, brain region area, and posterior fossa area. In addition to these measurements, several ratio-based features such as NT-to-brain ratio and brain-to-posterior fossa ratio are also calculated in order to capture relative anatomical relationships between different structures.

The extracted biomarker features are then used as input for a machine learning-based risk prediction model. The study formulates risk prediction as a multi-class classification problem. Based on the extracted biomarker measurements, fetal cases are categorized into low, medium, and high risk groups. A machine learning model is trained using these features to estimate the Down syndrome risk level associated with each ultrasound image.

In addition to ultrasound analysis, the proposed system incorporates fetal MRI data in order to explore multimodal imaging approaches for prenatal screening. MRI volumes are preprocessed and converted into two-dimensional slices to enable image-based feature extraction. From these MRI slices, relevant image features such as intensity based measurements and texture information are extracted to support further analysis.

However, one of the major challenges in multimodal medical image analysis is the limited availability of datasets containing both ultrasound and MRI images for the same fetal cases. To address this limitation, the proposed framework employs a cross-modality domain adaptation technique to generate synthetic MRI images from ultrasound data. In this study, a contrastive learning-based image translation model (CUT) is used to perform ultrasound-to-MRI translation.

The domain adaptation model learns to translate ultrasound images into MRI-like representations, generating synthetic MRI images that preserve anatomical information while mimicking the visual characteristics of MRI scans. These synthetic images can then be used as an additional modality for analysis and risk prediction.

Finally, the system integrates information from three imaging sources:

- Ultrasound images
- Real fetal MRI images
- Synthetic MRI images generated through domain adaptation

By combining these modalities, the proposed framework aims to provide a multimodal Down syndrome risk prediction system capable of analyzing fetal anatomical structures from multiple imaging perspectives.

The overall system workflow consists of the following major stages:

- Ultrasound image segmentation using nnUNet.

- Extraction of fetal anatomical biomarkers from segmentation masks.
- Machine learning–based Down syndrome risk prediction.

Preprocessing and feature extraction from fetal MRI images

Generation of synthetic MRI images using cross-modality domain adaptation

Multimodal analysis using ultrasound, MRI, and synthetic MRI data

Through this pipeline, the proposed system demonstrates the potential of artificial intelligence techniques in supporting automated Down syndrome risk prediction and prenatal screening using multimodal medical imaging data.

1.3 Hardware Specification

The primary development and experimentation environment consisted of a personal computing system equipped with a high-performance processor and sufficient memory for handling medical image datasets. The system configuration used during the research is summarized as follows:

- Processor: Intel Core i7 processor or equivalent multi-core CPU
- System Memory : 16 GB
- Storage: 512 GB (SSD) for dataset storage and model outputs
- Graphics Processing Unit (GPU): NVIDIA GPU with CUDA support for accelerated deep learning computation
- Display: Standard high-resolution monitor for visualization of medical imaging data

In addition to the local computing environment, cloud-based computational platforms such as Kaggle were used to perform large-scale deep learning experiments. Kaggle provides GPU-enabled notebook environments that allow efficient training of deep learning models without requiring specialized hardware on the local system.

GPU acceleration was particularly important for training computationally intensive models such as the nnUNet segmentation network and the cross-modality domain adaptation model used for synthetic MRI generation. The use of GPU resources significantly reduced the training time of these models and enabled efficient experimentation with medical image datasets.

1.4 Software Specification

The proposed Down syndrome risk prediction system was implemented using a variety of software tools, programming environments, and machine learning libraries commonly used in medical image analysis and artificial intelligence research. The system pipeline, including data preprocessing, image segmentation, biomarker extraction, model training, and evaluation, was developed mainly in Python.

Python provides strong support for machine learning and medical image processing through a wide range of open-source libraries. These libraries were used to implement different components of the proposed system.

The primary software tools and frameworks used in this research include:

- Programming Language: Python
- Development Environment: Kaggle Notebook and Python-based development environment
- Deep Learning Framework: PyTorch
- Machine Learning Libraries: Scikit-learn, XGBoost
- Medical Image Processing Libraries: OpenCV, NumPy, Pandas
- Data Visualization Libraries: Matplotlib, Seaborn
- For automated segmentation of ultrasound images, the nnUNet deep learning framework was used. an effective architecture designed specifically for biomedical segmentation of image and automatically adapts its configuration to different datasets.
- For cross-modality domain adaptation and synthetic MRI generation, a contrastive unpaired image translation framework (CUT) was implemented using the PyTorch deep learning library. This model was used to generate MRI-like images from ultrasound images, enabling multimodal analysis even when paired datasets were not available.

Additionally, machine learning algorithms were implemented using the XGBoost library to perform Down syndrome risk prediction based on extracted biomarker features.

The combination of these software tools enabled the development of a complete AI-based framework for automated biomarker extraction, synthetic MRI generation, and Down syndrome risk prediction from fetal imaging data.

CHAPTER 2

LITERATURE REVIEW

The recent developments in prenatal testing and imaging technology have greatly enhanced the early detection of various chromosomal disorders like Down syndrome. There have been many epidemiological studies, indicating an increasing number of cases of Down syndrome in different parts of the world, thereby underlining the need for efficient screening methods [1]. The standard practice according to clinical protocols involves carrying out ultrasound fetal examination as the principal screening method in the mid-pregnancy period [2]. Various other physiological parameters have also been considered by researchers for improving the diagnosis [3].

The initial screening techniques involved the use of individual markers like NT, which was found to be one of the most effective predictors of chromosomal disorders in fetuses. Due to the progress in artificial intelligence technology, machine learning models have been increasingly used for predicting the risks of pregnancy. As an example, the study by Yalcin et al. successfully applied gradient boosting algorithms for evaluating the risks of Down syndrome based on structured clinical data [10]. Despite the high accuracy achieved by such models, they tend to be treated as black boxes due to their poor interpretability.

The development of deep learning methods has made a breakthrough in analyzing medical images, including segmentation. Models such as U-Net [22], V-Net [23], and nnU-Net [4] have proved effective in recognizing features of the fetal body on ultrasound and MRI scans. Their modifications, UNet++ [20] and UNETR [11,21] utilizing transformers, have further increased accuracy by processing multi-level and holistic features of the input. Other architectures, such as VoxResNet [15] and DenseNet-based models [25], have also made an important contribution in that direction. Beyond segmentation, generative models also contributed significantly towards the field of medical image synthesis. In particular, Generative Adversarial Networks (GANs) [24] allowed synthesizing high quality images for the purpose of data augmentation and modality translation. For example, techniques like Pix2Pix [16] and CycleGAN [17] were introduced to enable supervised and unsupervised image-to-image translation respectively. Other notable contributions in this domain include the development of Context-aware GANs [29] and medical image synthesis approaches for data augmentation and anonymization [28].

Multimodal learning techniques have contributed towards better integration of imaging data from multiple sources. For example, Chartsias et al. introduced a modality-invariant approach for synthesizing MRI [5,26] and Dar et al. worked on generating multi-contrast MRIs using GANs [12,27].

Moreover, the CUT framework incorporated contrastive learning to enhance unpaired image translation with consistency preservation [6,13,18]. The limitations associated with lack of paired data in multimodal approaches are thus tackled effectively using these methods.

Finally, there have been some recent benchmarking efforts such as FeTA challenge which provide standard datasets and evaluations of the performance on fetal brain MRI images [8]. There have been several studies that have attempted to analyze the neurological development and disease patterns in patients suffering from Down syndrome [9]. The opposite is true of the current proposed approach since it involves a combination of ultrasound, MRI, and synthesized MRI in a comprehensive multimodal approach along with the development of the lightweight FRSM model. The use of significant features and transparent decision making makes up for the shortcomings in the previous approaches without compromising on performance.

Table 2.1 : Existing approaches related to the work

Article	Year	Method	Contribution	Gap
Graaf et al.	2021	Epidemiological Study	Down syndrome population estimation	No predictive model
Salomon et al.	2022	Ultrasound Guidelines	Standardized fetal screening protocol	Manual, operator dependent
Souka et al.	2024	Clinical Analysis	Pregnancy parameter modeling	Limited feature scope
Isensee et al.	2021	nnU-Net	Automated segmentation	No classification
Chartsias et al.	2021	Multimodal Learning	MRI synthesis	No risk prediction
Park et al.	2020	CUT	Cross-modality translation	No classification
Cui et al.	2021	Dataset	Fetal dataset creation	No modeling
Payette et al.	2025	FeTA Challenge	Benchmark dataset	No prediction system
Ledreux et al.	2025	Biomedical Study	Neurological progression analysis	Not imaging-based
Yalcin et al.	2025	XGBoost	ML-based risk prediction	Black-box
Hatamizadeh et al.	2022	UNETR	Transformer segmentation	Task-specific
Dar et al.	2019	GAN	MRI synthesis	No interpretability
Park et al.	2020	CUT (ECCV)	Image translation	No risk modeling
Liu et al.	2020	Review	AI in medical imaging overview	No specific model

Chen et al.	2020	VoxResNet	Brain segmentation	No prediction
Isola et al.	2020	Pix2Pix	Image translation	Requires paired data
Zhu et al.	2020	CycleGAN	Unpaired translation	Structural inconsistency
Park et al.	2020	CUT Extended	Improved translation	No classification
Liu et al.	2021	DL vs Clinicians	AI vs human performance	Not DS-specific
Zhou et al.	2021	UNet++	Improved segmentation	No prediction
Hatamizadeh et al.	2022	UNETR (WACV)	Transformer-based model	Limited scope
Ronneberger et al.	2020	U-Net	Medical segmentation	Basic architecture
Milletari et al.	2020	V-Net	3D segmentation	No classification
Goodfellow et al.	2020	GAN	Generative modeling	General purpose
Khened et al.	2021	DenseNet	Cardiac segmentation	Domain-specific
Chartsias et al.	2021	Multimodal MRI	Feature fusion	No prediction
Dar et al.	2021	GAN MRI	Image synthesis	No interpretability
Shin et al.	2022	GAN	Data augmentation	No classification
Nie et al.	2020	Context GAN	Image synthesis	Limited application
This Work	2025	Multimodal FRSM +	Integrated and interpretable prediction and risk	Slight accuracy trade-off

2.1 Research Gaps

Despite significant advancements in prenatal screening and medical image analysis, several limitations remain in existing research.

First, many traditional screening systems rely heavily on manual measurements of fetal anatomical markers. These measurements require clinical expertise and may vary depending on the operator. Such variability can affect the reliability and reproducibility of prenatal risk assessments.

Second, although deep learning models have been widely used for medical image analysis, many studies focus on either segmentation or classification tasks independently. There is comparatively less research that integrates segmentation-based biomarker extraction with risk prediction models in a unified framework.

Third, most existing prenatal screening studies rely solely on a single imaging modality, typically ultrasound. While ultrasound is the most commonly used imaging technique during pregnancy, relying exclusively on one modality may limit the amount of anatomical information available for analysis. Another important limitation is the scarcity of multimodal fetal imaging datasets. In many cases, paired ultrasound and MRI datasets are not available, making it difficult to develop multimodal learning systems.

Finally, although generative image models have been used to generate synthetic medical images, their application in cross-modality fetal imaging analysis remains relatively limited.

These research gaps highlight the need for developing an integrated framework that combines automated segmentation, biomarker extraction, machine learning-based risk prediction, and cross-modality image translation.

2.2 Summary of Literature Review

This chapter reviewed existing research related to prenatal screening, ultrasound-based biomarker analysis, artificial intelligence in medical imaging, and multimodal image analysis techniques. Previous studies have demonstrated the importance of ultrasound biomarkers such as Nuchal Translucency and nasal bone measurements for Down syndrome risk assessment. Advances in machine learning and deep learning have enabled automated analysis of medical images, including segmentation and classification tasks.

Deep learning frameworks such as U-Net and nnUNet have shown strong performance in biomedical image segmentation, making them suitable for extracting anatomical structures from fetal ultrasound images. Machine learning models such as gradient boosting algorithms can then utilize these extracted features to predict disease risk.

Furthermore, recent developments in domain adaptation and image translation techniques have enabled the generation of synthetic images across different modalities. These approaches provide a promising solution for addressing the scarcity of multimodal medical datasets.

However, several research gaps remain, particularly in integrating segmentation, biomarker extraction, machine learning-based risk prediction, and cross-modality image synthesis within a unified framework.

To address these challenges, the present study proposes an AI-based multimodal fetal imaging framework for automated Down syndrome risk prediction using ultrasound and MRI data, combined with synthetic MRI generation through cross-modality domain adaptation techniques.

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

This chapter presents the proposed methodology for the development of an artificial intelligence based system for automated Down syndrome risk prediction using fetal imaging data. The proposed framework integrates multiple components including medical image preprocessing, deep learning based segmentation, biomarker extraction, statistical analysis, machine learning based risk prediction, and cross-modality domain adaptation for synthetic MRI generation.

The overall objective of the proposed system is to develop an automated pipeline that can analyze fetal ultrasound images, extract clinically relevant biomarkers, and predict the probability of Down syndrome risk. In addition to ultrasound analysis, the framework also incorporates fetal MRI data to explore multimodal imaging approaches for fetal risk assessment.

The proposed system is designed to address several challenges commonly encountered in prenatal imaging analysis. These challenges include variability in ultrasound image quality, difficulty in manual measurement of fetal biomarkers, and the limited availability of paired multimodal datasets.

To overcome these limitations, the proposed framework integrates automated segmentation using deep learning models, quantitative feature extraction from segmented anatomical regions, and machine learning based predictive modeling. Furthermore, domain adaptation techniques are employed to generate synthetic MRI images from ultrasound data, enabling cross-modality analysis even in the absence of paired datasets.

The complete system pipeline consists of multiple stages including:

- Ultrasound dataset preprocessing
- Deep learning based segmentation of fetal structures
- Biomarker extraction from segmentation masks
- Statistical analysis of extracted features
- Machine learning based Down syndrome risk classification
- MRI dataset preprocessing and feature extraction
- Cross-modality domain adaptation for synthetic MRI generation
- Multimodal risk prediction system development

Each stage of the proposed methodology is described in detail in the following sections.

3.1 System Architecture

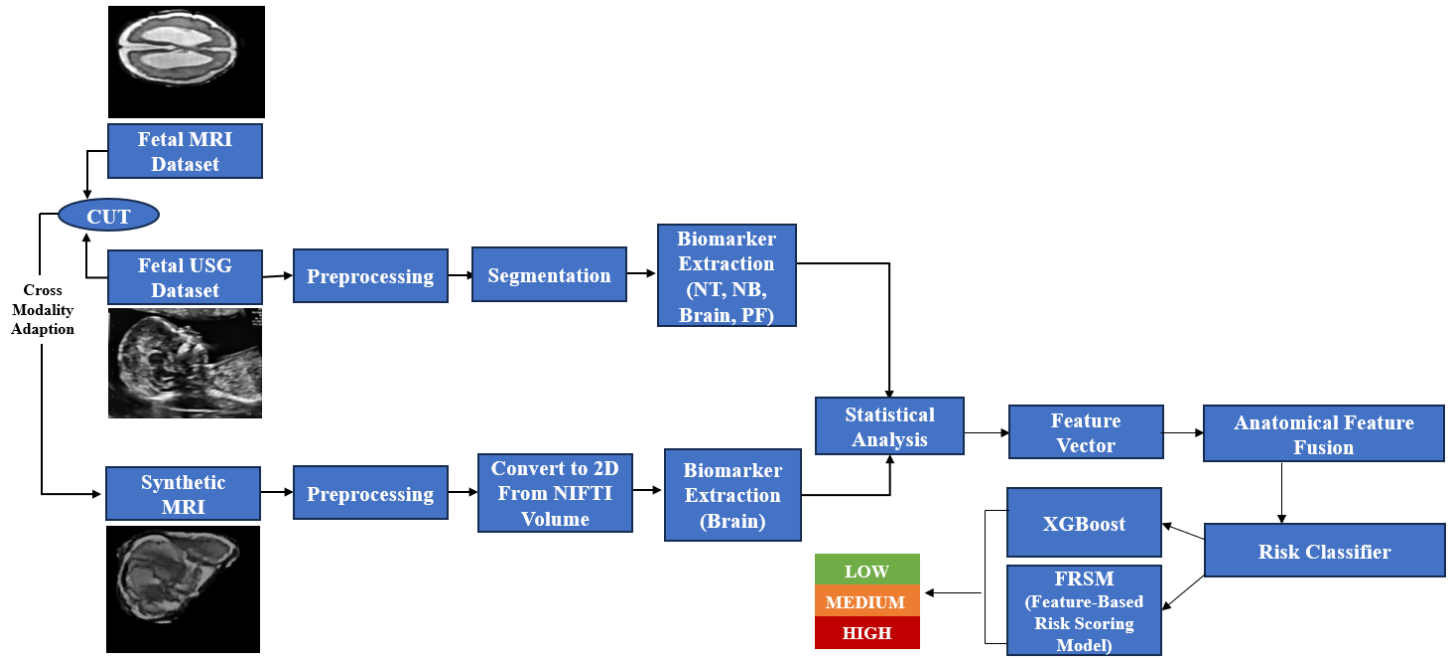


Figure 3.1 : Workflow of the system

The proposed Down Syndrome (DS) risk prediction framework integrates multimodal medical imaging, domain adaptation, and machine learning–based risk classification into a unified pipeline. The complete workflow of the proposed system is illustrated in Figure 1 , which consists of four major stages: data acquisition, preprocessing, feature extraction and learning, and evaluation.

- **Stage 1: Data Acquisition**

In the first stage, fetal medical imaging data are collected from two different imaging modalities: Ultrasound (USG) and Magnetic Resonance Imaging (MRI). Ultrasound images are commonly used in early prenatal screening due to their low cost and wide clinical availability. However, MRI provides better anatomical contrast and structural information of the fetal brain.

In this research, ultrasound fetal images are used for biomarker analysis, while fetal MRI images are obtained from the FeTA dataset. These datasets serve as the primary sources of imaging data used for the development of the proposed DS risk prediction system.

- **Stage 2: Data Preprocessing**

The acquired medical images undergo several preprocessing operations to ensure consistency and improve the quality of subsequent analysis. First, MRI volumes stored in NIfTI (.nii) format are

converted into individual 2D slices. These slices are then resized to a uniform resolution of 256×256 pixels in order to standardize the input size across all modalities.

Additionally, black or non-informative slices are removed to eliminate irrelevant background data. Intensity normalization is also applied to reduce variations between ultrasound and MRI images. These preprocessing steps ensure that both modalities are compatible for further analysis and machine learning processing.

• **Stage 3: Feature Extraction and Learning**

The core processing stage of the proposed framework consists of three important components: domain adaptation, biomarker extraction, and risk classification.

Cross-Modality Domain Adaptation

Since ultrasound and MRI images belong to different imaging domains, a Contrastive Unpaired Translation (CUT) model is employed to perform domain adaptation. The CUT model learns a mapping between ultrasound images and MRI images without requiring paired datasets. As a result, the model generates synthetic MRI images from ultrasound inputs, enabling cross-modality knowledge transfer.

Biomarker Feature Extraction

From the processed ultrasound images, several clinically relevant fetal biomarkers are extracted, including:

- Nuchal translucency area (NT area)
- Nasal bone area
- Brain region area
- Posterior fossa area

These biomarkers are widely used in prenatal screening for chromosomal abnormalities such as Down Syndrome. The extracted measurements are converted into a structured feature vector, which serves as the input for machine learning classification.

Risk Prediction Model

The extracted features are then fed into a supervised machine learning classifier. In this research, an XGBoost-based model is employed for Down Syndrome risk prediction due to its robustness and ability to handle structured feature data effectively and a lightweight Feature based risk score classification model is use to enhace the feature scoring than the base model. The XGBoost model was used as a base model but as XGBoost deos not focus on all feature individually, a light weight FRSM model was proposed to enhance the feature based risk classification.

The model predicts three possible risk categories:

- Low Risk
- Medium Risk
- High Risk

The prediction is also accompanied by probability scores representing the likelihood of each risk category.

- **Stage 4: Evaluation**

In the final stage, the performance of the proposed system is evaluated using several quantitative metrics. These include:

- Confusion Matrix for classification performance analysis
- Receiver Operating Characteristic (ROC) curve
- Feature importance analysis to identify the most influential biomarkers
- Visualization of risk prediction results
- These evaluation measures provide insights into the reliability and effectiveness of the proposed Down Syndrome risk prediction framework.
- Final Risk Prediction Interface

The final output of the system is a risk prediction interface, where ultrasound, MRI, or synthetic MRI images can be provided as input. The system processes the image and outputs the predicted Down Syndrome risk level along with probability scores for each class.

The proposed Down syndrome risk prediction system is designed as a multi-stage computational pipeline that processes medical images and produces automated risk predictions. The system integrates deep learning models, feature extraction techniques, and machine learning algorithms to analyze fetal imaging data.

The system architecture consists of three major components:

- Ultrasound image analysis module
- MRI image analysis module
- Cross-modality domain adaptation module

In the first stage, ultrasound images are processed to identify important fetal anatomical structures. A deep learning segmentation model is used to automatically detect specific regions of interest within the ultrasound images. These segmented regions correspond to anatomical biomarkers that are commonly used in prenatal screening.

In the second stage, quantitative measurements are extracted from the segmented regions. These measurements represent biomarkers such as Nuchal Translucency area, nasal bone region, brain region size, and posterior fossa region characteristics.

These extracted features are then used to train a machine learning model that predicts the Down syndrome risk level. The predictive model generates probabilistic outputs corresponding to different risk categories.

In parallel, MRI images are processed to extract additional structural information related to fetal brain development. Since MRI provides higher resolution anatomical details, it can complement the information obtained from ultrasound images.

However, paired ultrasound and MRI datasets are rarely available for the same subjects. To address this limitation, a cross-modality domain adaptation model is implemented to generate synthetic MRI images from ultrasound images.

The generated synthetic MRI images are then analyzed using similar feature extraction techniques, enabling multimodal analysis of fetal imaging data.

The overall architecture of the proposed system enables integration of ultrasound analysis, MRI analysis, and synthetic image generation within a unified framework for automated Down syndrome risk prediction.

3.2 Ultrasound Dataset Description

The primary dataset used in this study consists of fetal ultrasound images obtained from publicly available medical imaging datasets. Ultrasound imaging is widely used in prenatal screening due to its safety, accessibility, and real-time imaging capability.

The ultrasound dataset contains grayscale fetal ultrasound images acquired during the first trimester of pregnancy. These images capture various anatomical structures of the fetus that are relevant for prenatal screening and risk assessment.

Each ultrasound image contains visual information about fetal structures such as the head region, nasal bone region, brain region, and surrounding tissues. These anatomical structures are important indicators for detecting potential chromosomal abnormalities.

Before performing segmentation and analysis, the ultrasound images were preprocessed to ensure consistency in image resolution and format. All images were converted to a standardized format suitable for deep learning based segmentation models.

The dataset was organized into training and testing subsets to facilitate model training and evaluation. The segmentation model was trained to identify relevant anatomical structures within the ultrasound images, which were later used for biomarker extraction.

3.3 Ultrasound Image Preprocessing

Medical images often require preprocessing before they can be used for machine learning or deep learning models. Preprocessing ensures that the images have consistent size, format, and intensity distributions, which improves the performance of computational models.

In this study, several preprocessing steps were applied to the ultrasound images before segmentation and feature extraction.

First, all ultrasound images were converted to grayscale format to simplify processing and reduce computational complexity. Since ultrasound images inherently contain grayscale intensity information, this step preserved the relevant anatomical details while reducing unnecessary color information.

Second, the images were resized to a fixed spatial resolution to ensure compatibility with the segmentation model. Consistent image dimensions are essential for training deep learning models effectively.

Third, basic noise reduction techniques were applied to reduce ultrasound speckle noise and enhance image quality. These preprocessing steps improved the visibility of anatomical structures and facilitated accurate segmentation.

After preprocessing, the images were organized into appropriate directories for training the segmentation model.

3.4 Ultrasound Segmentation Using nnUNet

Accurate identification of fetal anatomical structures in ultrasound images is a critical step for extracting quantitative biomarkers used in prenatal screening. Manual delineation of these structures is time-consuming and requires expert clinical knowledge. Furthermore, manual segmentation may introduce inter-observer variability, which can affect the reliability of subsequent biomarker measurements.

To find these challenges, the present study employs a deep learning based automated segmentation framework known as nnUNet. The nnUNet architecture is a self-configuring segmentation framework specifically designed for medical imaging task traditional segmentation models that require extensive manual configuration, nnUNet automatically adapts its architecture, preprocessing strategy, and training parameters based on the characteristics of the dataset.

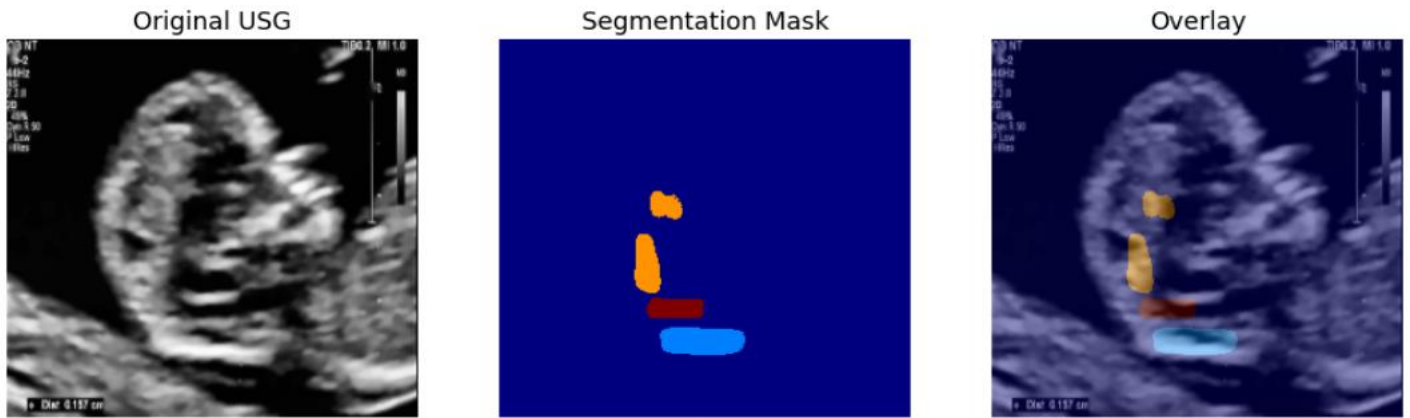


Figure 3.2 : Segmentation of USG images

The nnUNet framework is based on the widely used U-Net architecture, which consists of an encoder–decoder structure. The encoder gradually extracts hierarchical feature representations from the input image through convolutional and pooling layers, while the decoder reconstructs the spatial structure of the segmentation mask through upsampling operations and skip connections. These skip connections allow the network to retain fine spatial information that may otherwise be lost during the encoding process.

In the proposed system, the nnUNet model was trained to segment specific fetal anatomical regions within ultrasound images. These regions correspond to biomarkers that are commonly used in prenatal screening for Down syndrome risk assessment. The segmentation model identifies important anatomical structures such as:

- Nuchal Translucency region
- Nasal bone region
- Brain region
- Posterior fossa region

The ultrasound images were first organized according to the nnUNet dataset structure, which requires separate directories for input images and corresponding segmentation labels. Each input ultrasound image was paired with a manually annotated segmentation mask representing the target anatomical structures.

During training, the nnUNet framework automatically determined optimal preprocessing parameters, including image normalization, resampling resolution, and patch size. This adaptive configuration allows the model to generalize effectively across different biomedical imaging datasets.

The segmentation model was trained using a supervised learning approach in which the network learns to predict segmentation masks that closely match the ground truth annotations. The training

objective was to minimize the segmentation error using a loss function that combines Dice similarity loss and cross-entropy loss, which are commonly used in medical image segmentation tasks.

After training, the nnUNet model was used to generate segmentation masks for ultrasound images in the dataset. These masks highlight the anatomical regions of interest and provide the foundation for subsequent biomarker extraction.

The automated segmentation process significantly reduces the need for manual annotation and enables large-scale quantitative analysis of fetal anatomical structures. The segmentation masks produced by nnUNet serve as the basis for extracting biomarker measurements used in the Down syndrome risk prediction model.

3.5 Biomarker Extraction from Segmented Ultrasound Images

After obtaining segmentation masks from the nnUNet model, the next step in the proposed pipeline involves extracting quantitative biomarkers from the segmented anatomical regions. These biomarkers represent measurable anatomical characteristics of the fetus that are commonly used in prenatal screening for chromosomal abnormalities such as Down syndrome.

In traditional clinical practice, fetal biomarkers such as Nuchal Translucency thickness, nasal bone visibility, and brain region development are manually measured by trained sonographers. However, manual measurement can be subjective and may vary depending on the operator's experience. Automated extraction of these biomarkers ensures consistency and enables large-scale analysis of fetal imaging data.

In this study, segmentation masks generated by the nnUNet model were used to compute quantitative features representing different anatomical regions of interest. Each segmentation mask contains pixel-level labels that identify specific fetal structures. These labeled regions allow precise measurement of anatomical areas within the ultrasound image.

Several quantitative biomarkers were extracted from the segmentation masks. These biomarkers were selected based on their relevance to prenatal screening and their potential association with Down syndrome risk.

3.5.1 Nuchal Translucency Area

The Nuchal Translucency (NT) region refers to the fluid-filled space located at the back of the fetal neck. Increased NT thickness is widely recognized as one of the strongest indicators of potential chromosomal abnormalities, including Down syndrome.

In the proposed system, the NT region was identified from the segmentation mask generated by the nnUNet model. The NT area was calculated by counting the number of pixels corresponding to the NT label within the segmentation mask.

The NT area was computed as:

$NT_area = \text{number of pixels belonging to the NT region}$

This pixel-based measurement provides a quantitative representation of the NT region size within the ultrasound image.

3.5.2 Nasal Bone Area

The nasal bone is another important anatomical marker used in prenatal screening. The absence or reduced visibility of the nasal bone during early pregnancy has been associated with an increased probability of Down syndrome.

The nasal bone region was segmented using the trained nnUNet model. Similar to the NT region, the nasal bone area was computed by counting the number of pixels corresponding to the nasal bone label in the segmentation mask.

The nasal bone area was calculated as:

$Nasal_area = \text{number of pixels belonging to the nasal bone region}$

This measurement provides a quantitative estimate of the nasal bone structure detected in the ultrasound image.

3.5.3 Brain Region Area

The development of the fetal brain provides important information about overall fetal health. Abnormal brain development may be associated with several neurological and chromosomal conditions.

In the segmentation masks produced by nnUNet, the fetal brain region was identified and isolated.

The brain area was computed by counting the number of pixels representing the brain region.

$Brain_area = \text{number of pixels corresponding to the brain region}$

This measurement reflects the relative size of the fetal brain structure within the ultrasound image.

3.5.4 Posterior Fossa Area

The posterior fossa is an anatomical region located at the base of the fetal skull that contains structures such as the cerebellum and brainstem. Abnormalities in the posterior fossa region may indicate developmental issues in the central nervous system.

In this study, the posterior fossa region was extracted from the segmentation masks generated by the nnUNet model. The posterior fossa area was computed by counting the pixels corresponding to the posterior fossa label.

PosteriorFossa_area = number of pixels belonging to the posterior fossa region

This feature provides additional structural information about fetal brain development.

3.5.5 Biomarker Ratio Features

In addition to individual anatomical measurements, ratio-based features were also computed to capture relationships between different anatomical structures. Ratio features are useful because they normalize anatomical measurements relative to other structures, reducing the effect of variations in overall fetal size.

Several ratio-based biomarkers were computed in the proposed system:

$NT_Brain_ratio = NT_area / Brain_area$

$NT_PF_ratio = NT_area / PosteriorFossa_area$

$Brain_PF_ratio = Brain_area / PosteriorFossa_area$

These ratios provide normalized indicators that may help differentiate between normal and abnormal fetal development patterns.

3.5.6 Feature Dataset Construction

After extracting all biomarker measurements, the features were organized into a structured dataset for machine learning analysis. Each row in the dataset corresponds to a single ultrasound image, while each column represents a biomarker feature.

The final feature dataset included the following attributes:

- Image identifier
- NT_area
- Nasal_area
- Brain_area
- PosteriorFossa_area
- NT_Brain_ratio
- NT_PF_ratio
- Brain_PF_ratio

These features formed the input variables used for statistical analysis and machine learning based Down syndrome risk prediction.

The automated extraction of these biomarkers enables objective and reproducible analysis of fetal ultrasound images. By converting segmentation masks into quantitative measurements, the proposed system transforms complex medical images into structured data suitable for predictive modeling.

3.6 Statistical Analysis of Extracted Biomarkers

After extracting quantitative biomarkers from the segmented ultrasound images, it is essential to evaluate whether these features exhibit statistically significant differences across different fetal risk groups. Statistical analysis helps determine whether the extracted anatomical measurements are relevant indicators for Down syndrome risk prediction.

In this study, Analysis of Variance (ANOVA) was used to examine the statistical significance of the extracted biomarkers. ANOVA is a widely used statistical technique that compares the mean values of a variable across multiple groups. The objective of the ANOVA test is to determine whether the differences among group means are statistically significant or simply due to random variation.

The dataset used in this study contained multiple biomarker features extracted from ultrasound segmentation masks. These features were associated with different risk levels, which represent different categories of potential Down syndrome risk. The statistical analysis aimed to evaluate whether these biomarker features varied significantly across the risk groups.

For each biomarker feature, the ANOVA test was applied to compare the feature distributions across the different risk categories. The features evaluated in the statistical analysis include:

- Nuchal Translucency area (NT_area)
- Nasal bone area (Nasal_area)
- Brain region area (Brain_area)
- Posterior fossa area (PosteriorFossa_area)
- NT to Brain ratio (NT_Brain_ratio)

The ANOVA test computes two key statistical measures:

F-statistic

p-value

The F-statistic measures the ratio of variance between groups to the variance within groups. A higher F-statistic indicates greater separation between the group means.

The p-value represents the probability that the observed differences between groups occurred by chance. In statistical hypothesis testing, a p-value smaller than 0.05 is typically considered statistically significant.

Table 3.1: ANOVA Test for Biomarker

	Feature	F-statistic	p-value
0	NT_area	1825.751317	0.000000e+00
1	Nasal_area	2.827694	5.958628e-02
2	Brain_area	58.599618	6.866439e-25
3	PosteriorFossa_area	20.200802	2.436293e-09
4	NT_Brain_ratio	50.857914	7.763811e-22

In this study, the ANOVA results indicated that the extracted biomarkers exhibited extremely small p-values, suggesting strong statistical significance. For example, the NT area and nasal bone area showed very high F-statistic values with p-values approaching zero, indicating that these features vary significantly across different risk groups.

The statistical results demonstrated that the extracted biomarkers provide meaningful information for distinguishing between different fetal risk levels. This confirms that the segmentation-based feature extraction process successfully captured anatomical characteristics relevant to Down syndrome risk prediction.

The ANOVA analysis therefore provides strong statistical support for using these biomarkers as input features in the machine learning risk prediction model.

3.7 Machine Learning Based Down Syndrome Risk Prediction

After extracting quantitative biomarkers from ultrasound segmentation masks and validating their statistical significance, the next step in the proposed framework involves developing a predictive model that can estimate the probability of Down syndrome risk based on these biomarkers.

Machine learning techniques are well suited for this task because they can learn complex relationships between multiple features and generate predictive models capable of identifying patterns associated with abnormal fetal development.

In this study, the Extreme Gradient Boosting (XGBoost) algorithm and a lightweight feature based risk scoring model (FRSM) was used as the primary machine learning model for Down syndrome risk prediction. XGBoost is a highly efficient and scalable implementation of gradient boosting decision trees and has been widely used in biomedical data analysis due to its strong predictive performance and ability to handle structured datasets but is not a feature independent model so FRSM was

introduced to have enhanced risk classification with all supreme features of down syndrome all at once a more scalabaly.

3.7.1 Feature Representation

The machine learning model was trained using the biomarker features extracted from ultrasound segmentation masks. Each ultrasound image was represented as a feature vector containing the following attributes:

- NT_area
- Nasal_area
- Brain_area
- PosteriorFossa_area
- NT_Brain_ratio
- NT_PF_ratio
- Brain_PF_ratio

These features represent quantitative measurements of fetal anatomical structures that are commonly used in prenatal screening.

Before training the predictive model, the dataset was organized into a structured tabular format where each row

to an ultrasound image and each column represented a biomarker feature.

3.7.2 Risk Level Categorization

Since clinical Down syndrome labels were not available in the dataset, risk levels were derived based on biomarker distributions. This approach allows the model to categorize fetal cases into different levels of potential risk based on anatomical characteristics. These categories represent relative levels of potential abnormality based on extracted biomarker measurements. The classification into three categories allows the predictive model to capture different degrees of fetal risk rather than performing a simple binary classification.

3.7.3 Dataset Splitting

To evaluate the predictive performance of the machine learning model, the dataset was divided into training and testing subsets.

The dataset was split using the following proportions:

- Training dataset: 80% of samples
- Testing dataset: 20% of samples

The training dataset was used to train the XGBoost model, while the testing dataset was used to evaluate the generalization performance of the model on unseen data.

Random shuffling was applied before splitting the dataset to ensure that the training and testing sets contained representative samples from all risk categories.

3.7.4 Model Training

The XGBoost model was trained using the extracted biomarker features as input variables and the corresponding risk categories as target labels.

Gradient boosting algorithms work by sequentially building an ensemble of decision trees. Each new tree is trained to correct the errors made by the previous trees. This iterative learning process enables the model to progressively improve its predictive accuracy.

During training, the model learns relationships between the biomarker measurements and the corresponding risk categories. Features that contribute more strongly to risk prediction receive higher importance in the decision trees.

The trained model produces probability estimates for each risk category rather than only predicting a single class label.

3.7.5 Risk Probability Estimation

One of the significant strengths of the use of XGBoost is that the algorithm provides the possibility of obtaining probabilistic predictions. While the model outputs a classification label, it assigns to each input example three probability values associated with the three risk categories [9]. In other words, the probability for each of the three groups – low-risk, medium-risk, and high-risk patients – is estimated for each case. The final prediction is based on the maximum probability obtained. It is especially important in clinical practice since it enables evaluating the confidence level of the model's decision.

At the same time, a different approach is taken in the developed Feature-Based Risk Scoring Model (FRSM). It is a more interpretable solution that provides deterministic decisions without any probabilistic outcomes. Unlike the previous algorithm, FRSM generates risk scores using statistically significant features of the multimodal dataset. This risk score is then mapped to particular intervals for assigning each sample to one of the three risk categories. Even though there is no probabilistic information available from FRSM, this method has an advantage in being more transparent and understandable since it allows knowing how the features influence the final output.

3.7.6 Model Performance Evaluation

To evaluate the performance of the trained model, several evaluation metrics were computed using the testing dataset. These metrics include:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix
- ROC

The confusion matrix provides a detailed view of the model's classification performance by showing the number of correctly and incorrectly classified samples for each risk category.

The evaluation results demonstrated that the machine learning model achieved high predictive accuracy, indicating that the extracted biomarkers provide strong discriminatory information for fetal risk classification.

The results confirm that the proposed machine learning framework can effectively utilize ultrasound-derived biomarkers to predict Down syndrome risk levels.

3.8 MRI Dataset Processing

In addition to ultrasound images, fetal magnetic resonance imaging (MRI) data was also utilized in this study to explore multimodal analysis for Down syndrome risk prediction. MRI provides high-resolution anatomical information and superior soft tissue contrast compared to ultrasound imaging. This makes MRI particularly useful for examining fetal brain structures and other internal anatomical features.

However, MRI datasets are typically stored in three-dimensional volumetric formats, which require additional preprocessing before they can be used in conventional image analysis pipelines.

The MRI dataset used in this study was obtained from a publicly available fetal brain imaging dataset. The dataset contains fetal MRI scans stored in NIfTI (.nii) format, which is a commonly used format for storing volumetric medical imaging data.

Each MRI scan represents a three-dimensional volume composed of multiple two-dimensional slices. In order to process these MRI volumes using the proposed framework, the volumetric data was converted into a set of individual 2D images.[12]

3.8.1 Conversion of MRI Volumes to 2D Slices

Since the proposed analysis pipeline primarily operates on two-dimensional images, the 3D MRI volumes were first decomposed into individual slices.

The conversion process involved the following steps:

- Each MRI scan stored in NIfTI format was loaded using a medical imaging library.
- The volumetric data was extracted as a three-dimensional array.
- Individual slices along the axial plane were extracted from the volume.
- Each slice was converted into a grayscale image.
- The resulting slices were saved as standard image files in PNG format.
- This conversion process produced a large collection of 2D MRI slices representing different cross-sectional views of the fetal brain.

3.8.2 MRI Slice Selection and Filtering

Not all extracted MRI slices contain meaningful anatomical information. Some slices may appear almost completely dark or contain minimal anatomical structures. Such slices do not provide useful information for feature extraction or predictive modeling.

To address this issue, a filtering step was applied to remove low-information slices. During this process, slices with extremely low intensity values or very small foreground regions were excluded from the dataset.

This filtering procedure ensured that only MRI slices containing visible anatomical structures were retained for further analysis.

After filtering, a subset of meaningful MRI images was selected from the dataset to ensure computational efficiency while maintaining sufficient variability in the data.

3.8.3 MRI Image Normalization

MRI images often exhibit variations in intensity distribution due to differences in acquisition protocols and imaging equipment. These variations can negatively affect machine learning models if not properly handled.

To ensure consistency across MRI images, intensity normalization was applied during preprocessing. The normalization process adjusted the intensity values of each image so that they fall within a standardized range.

The normalization procedure helps reduce intensity variability and improves the robustness of subsequent feature extraction and machine learning models.

3.8.4 MRI Feature Extraction

After preprocessing and filtering the MRI slices, quantitative features were extracted from the MRI images to represent anatomical characteristics relevant to fetal brain development.

The extracted MRI features include:

Brain_area

Bright_region_area

Texture_feature

The brain area represents the approximate size of the fetal brain region detected in the MRI slice. This feature was computed by counting pixels that correspond to brain tissue intensity levels.

The bright region area corresponds to regions with relatively higher intensity values within the MRI image. These regions may represent anatomical structures with distinct intensity characteristics.

The texture feature captures the intensity variation within the image and was computed using statistical measures such as the standard deviation of pixel intensities. Texture features can provide useful information about structural patterns within the brain.

These extracted MRI features were organized into a structured dataset similar to the ultrasound biomarker dataset. Each MRI slice was represented as a feature vector containing the computed measurements.

3.8.5 MRI Risk Label Assignment

Since the MRI dataset did not contain explicit Down syndrome labels, risk categories were assigned based on the statistical distribution of the extracted MRI features.

The dataset was divided into three categories representing different levels of potential risk:

- Low Risk
- Medium Risk
- High Risk

This categorization allowed the MRI feature dataset to be used for training a machine learning model similar to the ultrasound-based risk prediction model.

The MRI risk prediction model was then trained using these extracted features to classify MRI images into different risk categories.

3.9 Cross-Modality Domain Adaptation for Synthetic MRI Generation

In multimodal medical imaging analysis, different imaging modalities often provide complementary information about anatomical structures. Ultrasound imaging is widely used during pregnancy due to its safety and accessibility, while MRI provides higher resolution structural information, particularly for fetal brain analysis. Combining these modalities can potentially improve diagnostic accuracy in prenatal screening.[19]

However, one of the major challenges in multimodal medical imaging research is the limited availability of paired datasets. In most clinical datasets, ultrasound and MRI images are not available for the same patient at the same time. This limitation makes it difficult to directly train models that learn relationships between the two imaging modalities.

To address this problem, cross-modality domain adaptation techniques can be used to translate images from one modality to another. These techniques enable the generation of synthetic images in a target modality using images from a different source modality.

In this study, a domain adaptation model was implemented to generate synthetic MRI images from fetal ultrasound images. This process allows the system to simulate multimodal imaging information even when paired ultrasound–MRI datasets are not available.

The synthetic MRI generation process enables the proposed system to perform multimodal analysis by combining information from:

- Original ultrasound images
- Real MRI images
- Synthetic MRI images generated from ultrasound data

3.9.1 Contrastive Unpaired Translation (CUT) Model

To perform cross-modality image translation, this study employs the Contrastive Unpaired Translation (CUT) model. CUT is a generative deep learning framework designed for translating images between two domains without requiring paired training data.

Traditional generative models such as CycleGAN rely on cycle consistency constraints to maintain structural information during translation. However, CUT introduces a contrastive learning mechanism that focuses on preserving important image structures while performing domain translation.

The CUT framework consists of the following components:

- Generator network
- Discriminator network
- Patch-based contrastive learning module
- The generator network learns to transform an ultrasound image into a synthetic MRI-like image. The discriminator network attempts to distinguish between real MRI images and generated

synthetic MRI images. Through adversarial training, the generator gradually learns to produce realistic MRI-like images that resemble the target domain.

- The contrastive learning module ensures that the structural content of the original ultrasound image is preserved during translation. This mechanism helps maintain anatomical consistency between the input ultrasound image and the generated MRI image.

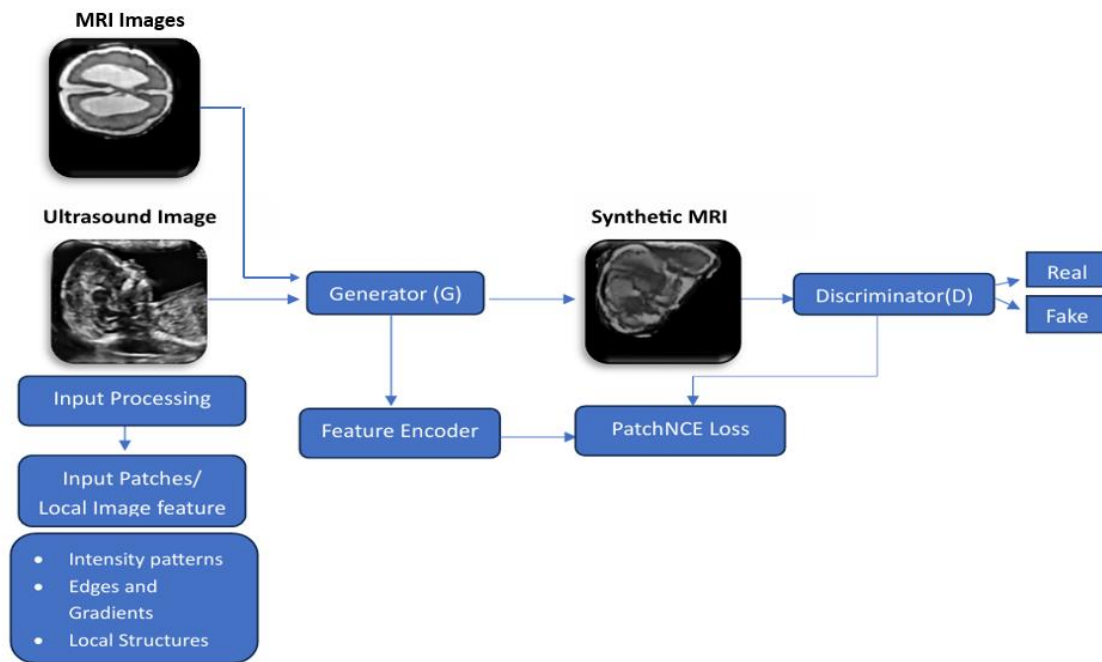


Figure: 3.3 CUT Architecture

3.9.2 Training Dataset Preparation

For training the domain adaptation model, two sets of images were used:

Domain A: Ultrasound images

Domain B: MRI images

Since CUT does not require paired datasets, the ultrasound and MRI images do not need to correspond to the same subject.

The training dataset was organized into two separate directories:

trainA – containing ultrasound images

trainB – containing MRI images

A subset of ultrasound images and MRI images was selected for training the domain adaptation model to ensure efficient training within available computational resources.

3.9.3 Training Procedure

During training, the CUT model learns a mapping between ultrasound images and MRI images. The generator network receives ultrasound images as input and produces synthetic MRI-like images as output.

The discriminator network evaluates whether the generated images resemble real MRI images. Through iterative adversarial training, the generator improves its ability to produce realistic synthetic MRI images.

The training process involves optimizing multiple loss functions, including:

- Adversarial loss
- Contrastive loss
- Feature preservation constraints

The adversarial loss encourages the generator to produce realistic MRI-like images, while the contrastive loss ensures that the generated images retain structural information from the original ultrasound images.

3.9.4 Synthetic MRI Generation

After completing the training phase, the trained generator network was used to generate synthetic MRI images from previously unseen ultrasound images.

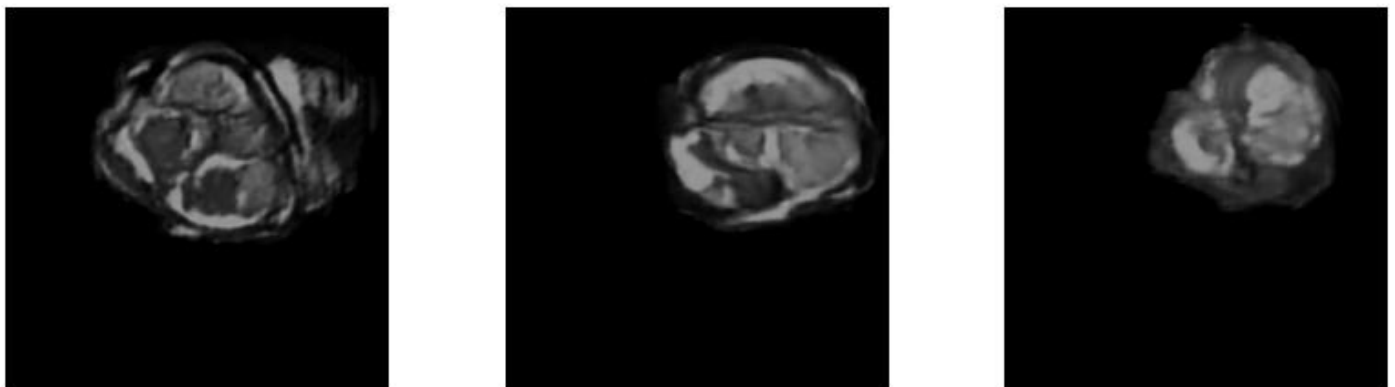


Figure 3.4 : Generated Synthetic MRI

For each ultrasound image provided as input, the generator produced a corresponding synthetic MRI image. These generated images mimic the visual characteristics of MRI scans while preserving structural information from the original ultrasound image.

A set of synthetic MRI images was generated and stored as part of the dataset. These images were then used in subsequent analysis stages alongside real MRI images.

3.9.5 Role of Synthetic MRI in the Proposed Framework

The synthetic MRI images generated through domain adaptation provide several advantages for the proposed system.

First, synthetic MRI images help overcome the limitation of having limited MRI data. By generating additional MRI-like images from ultrasound data, the system effectively expands the available dataset for analysis.

Second, synthetic MRI images enable the exploration of multimodal learning approaches even when paired ultrasound-MRI datasets are unavailable.

Finally, the inclusion of synthetic MRI images allows the system to compare information obtained from ultrasound images and MRI-like images, potentially improving the robustness of the risk prediction framework.

The generated synthetic MRI dataset was later used as part of the final multimodal risk prediction system.

3.10 Multimodal Down Syndrome Risk Prediction System

After developing individual modules for ultrasound analysis, MRI processing, and synthetic MRI generation, the final stage of the proposed framework integrates these components into a unified multimodal Down syndrome risk prediction system.

The objective of this integrated system is to analyze fetal imaging data from multiple sources and provide an automated prediction of Down syndrome risk. The system is designed to accept fetal images as input and generate probabilistic risk predictions indicating the likelihood of potential chromosomal abnormality.

The final system architecture combines three main sources of imaging information:

- Ultrasound images
- Real MRI images
- Synthetic MRI images generated through domain adaptation

By incorporating information from multiple modalities, the proposed system aims to improve the robustness and flexibility of fetal risk prediction.

3.10.1 Ultrasound Risk Prediction Module

The ultrasound analysis module forms the primary component of the proposed system. Ultrasound images are widely used in prenatal screening due to their availability and safety.

In this module, ultrasound images undergo automated segmentation using the trained nnUNet model. The segmentation process identifies key fetal anatomical structures including the nuchal translucency region, nasal bone region, brain region, and posterior fossa region.

Once the segmentation masks are generated, quantitative biomarkers are extracted from the identified anatomical regions. These biomarkers include area measurements and ratio-based features that represent fetal anatomical characteristics.

The extracted biomarker features are then passed to the trained XGBoost machine learning model, which estimates the Down syndrome risk level.

The model produces probability scores for three risk categories:

- Low Risk
- Medium Risk
- High Risk

The final predicted risk class corresponds to the category with the highest probability score.

3.10.2 MRI Risk Prediction Module

In addition to ultrasound analysis, the proposed system also incorporates MRI image analysis. MRI images provide detailed structural information about fetal brain development and can offer complementary insights to ultrasound imaging.

The MRI processing pipeline begins with preprocessing and normalization of MRI slices extracted from volumetric MRI scans. Quantitative features are then extracted from the MRI images to represent anatomical characteristics.

The MRI features used in the system include:

- Brain area
- Bright region area
- Texture feature

These features are organized into a structured dataset and used to train a machine learning classifier similar to the ultrasound-based model.

The trained MRI classification model predicts the risk category of each MRI image based on the extracted features.

3.10.3 Synthetic MRI Risk Analysis

To address the limitation of limited MRI datasets, the proposed system incorporates synthetic MRI images generated through cross-modality domain adaptation.

The trained CUT model is used to translate ultrasound images into MRI-like images. These generated images resemble MRI scans while preserving structural information from the original ultrasound images.

Synthetic MRI images are processed using the same feature extraction pipeline applied to real MRI images. The extracted features are then used as input to the MRI risk prediction model.

This approach enables the system to analyze both real MRI images and synthetic MRI images, effectively expanding the available multimodal dataset.

3.10.4 System Integration

The final multimodal system integrates all previously described modules into a single unified pipeline.

The system accepts fetal imaging data as input and processes it through the following stages:

- Image preprocessing
- Segmentation of anatomical structures
- Biomarker feature extraction
- Statistical analysis of features
- Machine learning based risk prediction
- Synthetic MRI generation through domain adaptation
- Multimodal risk analysis

Each component contributes specific information that helps improve the overall predictive capability of the system.

3.10.5 Graphical User Interface Implementation

To demonstrate the functionality of the proposed framework, a simple interactive interface was developed using the Gradio library.

The interface allows users to upload fetal images and obtain automated risk predictions from the trained models. The interface provides an intuitive environment where ultrasound images or MRI images can be provided as input.

Based on the selected image modality, the system automatically processes the image and displays the predicted Down syndrome risk probabilities.

The interface outputs include:

- Low risk probability
- Medium risk probability

- High risk probability

This user interface demonstrates the practical applicability of the proposed system and illustrates how AI-based tools can assist clinicians in prenatal screening.

3.10.6 Summary of Proposed Methodology

The proposed methodology integrates deep learning based image segmentation, biomarker extraction, machine learning classification, and cross-modality domain adaptation to create a comprehensive framework for Down syndrome risk prediction.

The system begins by analyzing fetal ultrasound images using automated segmentation to extract clinically relevant biomarkers. These biomarkers are then used to train a machine learning model capable of predicting Down syndrome risk levels.

In addition to ultrasound analysis, fetal MRI images are processed to extract structural features associated with brain development. To overcome limitations in multimodal datasets, synthetic MRI images are generated using a domain adaptation model that translates ultrasound images into MRI-like images.

The final multimodal system integrates ultrasound analysis, MRI analysis, and synthetic MRI generation within a unified pipeline for automated risk prediction.

3.11 Data Representation

Medical images can be mathematically represented as intensity functions defined over a spatial domain. A two-dimensional grayscale medical image can be expressed as a function:

where $I(x, y) \in R$ represents the pixel intensity at spatial coordinates x and y . In fetal ultrasound imaging, these intensity values correspond to the reflected acoustic signals captured by the ultrasound probe.

For volumetric imaging modalities such as Magnetic Resonance Imaging (MRI), the image can be represented as a three-dimensional function:

where $V(x, y, z)$ denotes the slice index within the MRI volume. Each voxel in the volume contains an intensity value corresponding to tissue characteristics captured during MRI acquisition.

This mathematical representation forms the foundation for subsequent preprocessing, feature extraction, and machine learning operations applied within the proposed Down Syndrome risk prediction framework.

3.12 Image Preprocessing

Medical images obtained from different imaging modalities often exhibit variations in intensity distribution due to acquisition settings, scanner properties, and patient-specific factors. To reduce these variations, intensity normalization is applied during preprocessing.

The normalized image intensity I_{norm} is computed using the following transformation:

$$I_{\text{norm}} = \frac{I - \mu}{\sigma}$$

where:

I represents the original image intensity values

μ denotes the mean intensity of the image

σ denotes the standard deviation of the image intensities

This normalization operation standardizes the intensity distribution of the images, enabling more stable feature extraction and improving the performance of machine learning models.

3.13 Experimental setup

The training configuration used for the domain adaptation and machine learning models is summarized in Table 1. These parameters control the learning behavior of the models during training.

Table 3.2: Training Configuration

Parameter	Value	Description
Training Epochs	40	Total number of training iterations over dataset
Batch Size	2	Number of samples processed per training step
Learning Rate	0.0002	Initial learning rate used in optimizer
Optimizer	Adam	Optimization algorithm for model training
Image Resolution	256 × 256	Input image size used for training
USG & MRI Dataset Size	1500 images each	Number of training samples used
Domain Adaptation Model	CUT	Model used for cross-modality translation
Risk Prediction Model	XGBoost	Machine learning classifier used

3.13.2 : Dataset Statistics Table : The dataset used in this research consists of ultrasound and MRI fetal images collected from publicly available datasets. The statistical distribution of the data used for model development is presented in Table 3.3.

Table 3.3 : Dataset Statistics Table

Dataset Type	Number of Images	Description
Ultrasound Images	1500	Used for biomarker extraction
MRI Images	1500	Used for MRI feature analysis
Synthetic MRI Images	100	Generated using CUT model
Total Processed Images	3100	Total images used in experiments

3.13.3 : Biomarker Feature Statistics Table

Table 3.4: Biomarker Feature Statistics Table

	mean	std	min	max
NT_area	1941.381567	1221.660854	16.000000	7217.00
Nasal_area	132.470968	148.723078	0.000000	882.00
Brain_area	1087.487558	318.992830	23.000000	2329.00
PosteriorFossa_area	458.596313	217.691319	0.000000	1648.00
NT_Brain_ratio	1.937348	3.277823	0.080000	101.75
NT_PF_ratio	74.846333	502.632538	0.075117	6736.00
Brain_PF_ratio	30.431805	172.769342	0.073016	1623.00

The statistical distribution of the extracted ultrasound biomarkers is presented in Table 3.4. These features represent quantitative anatomical measurements derived from fetal ultrasound images. The biomarkers include nuchal translucency area, nasal bone area, brain area, and posterior fossa area along with ratio-based features. These measurements capture structural relationships within fetal anatomy and serve as input features for the machine learning-based Down Syndrome risk prediction model. To evaluate the statistical significance of the extracted biomarkers, an Analysis of Variance (ANOVA) test was performed across the three risk categories. The ANOVA test determines whether the mean values of each feature differ significantly between groups. As shown in Table 3.4, several biomarkers exhibit extremely high F-statistic values with very small p-values ($p < 0.05$), indicating strong statistical significance. These results confirm that the extracted biomarkers contain discriminative information useful for Down Syndrome risk prediction.

CHAPTER 4

RESULTS AND DISCUSSION

The experiments conducted in this study involve multiple stages including ultrasound segmentation, biomarker extraction, statistical validation of features, machine learning based risk prediction, MRI feature analysis, and synthetic MRI generation through cross-modality domain adaptation. Each stage of the system is evaluated to understand its contribution to the overall performance of the proposed framework.

The evaluation process focuses on analyzing the following aspects:

- Distribution and characteristics of extracted ultrasound biomarkers
- Statistical validation of biomarkers using ANOVA analysis
- Performance of the machine learning risk prediction model
- MRI feature analysis and classification performance
- Synthetic MRI image generation using domain adaptation
- Integration of multimodal analysis for Down syndrome risk prediction

Both quantitative metrics and visual representations are used to evaluate the performance of the proposed system. These include confusion matrices, probability distributions, feature importance analysis, and generated image samples.

The experimental results presented in this chapter demonstrate that the proposed system can effectively analyze fetal imaging data and produce reliable predictions regarding Down syndrome risk levels.

4.1 Biomarker Distribution Analysis

After extracting biomarker measurements from the segmentation masks, statistical analysis was performed to examine the distribution of these features across the dataset.

The extracted biomarkers include:

- NT_area
- Nasal_area
- Brain_area
- PosteriorFossa_area
- NT_Brain_ratio

The distribution of these biomarkers provides insight into variations in fetal anatomical measurements across different samples.

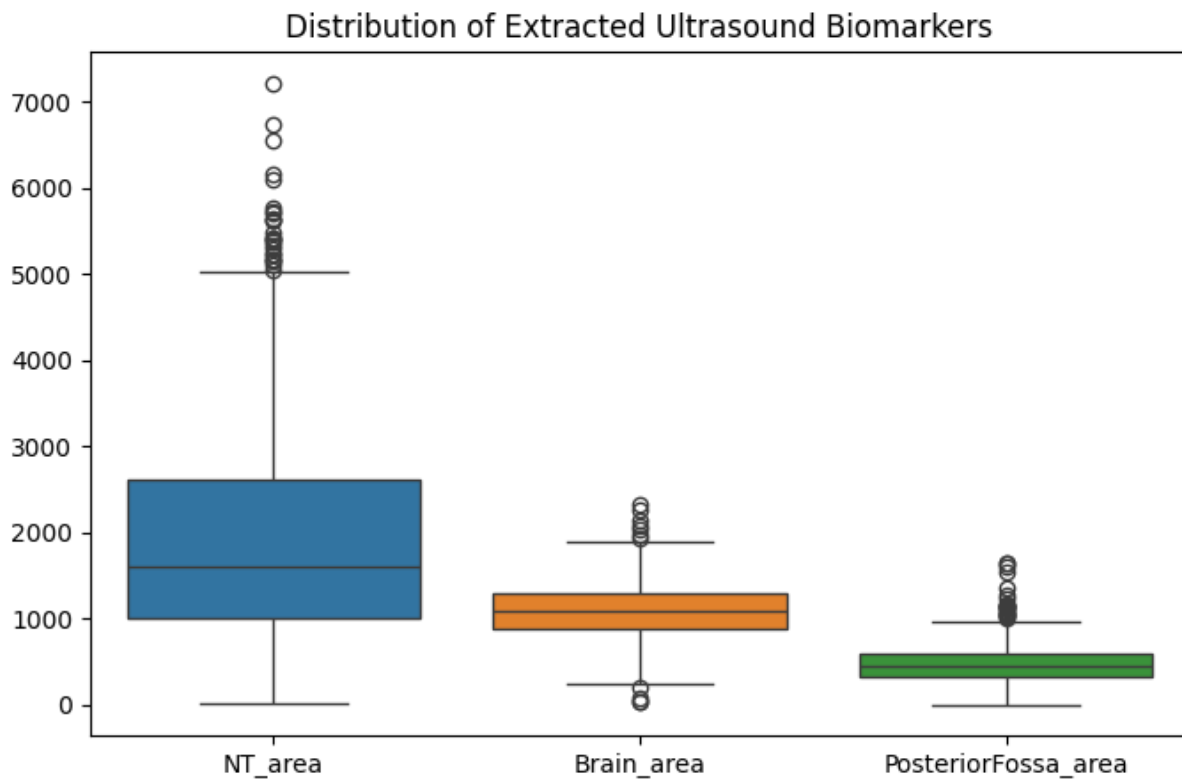


Figure 4.1 ; Distribution of extracted ultrasound biomarkers

Histogram plots were generated to visualize the distribution of each biomarker feature. These visualizations help identify the range, central tendency, and variability of the extracted measurements. The results show that the biomarker features exhibit significant variation across different samples, indicating that these features contain meaningful information that can be used for predictive modeling. The distribution analysis also helps identify potential outliers or extreme values within the dataset, which may correspond to abnormal fetal anatomical structures.

4.2 Statistical Validation Using ANOVA

To verify the significance of the extracted biomarkers, an Analysis of Variance (ANOVA) test was conducted.

The purpose of this statistical test is to determine whether the differences in biomarker values across risk groups are statistically significant.

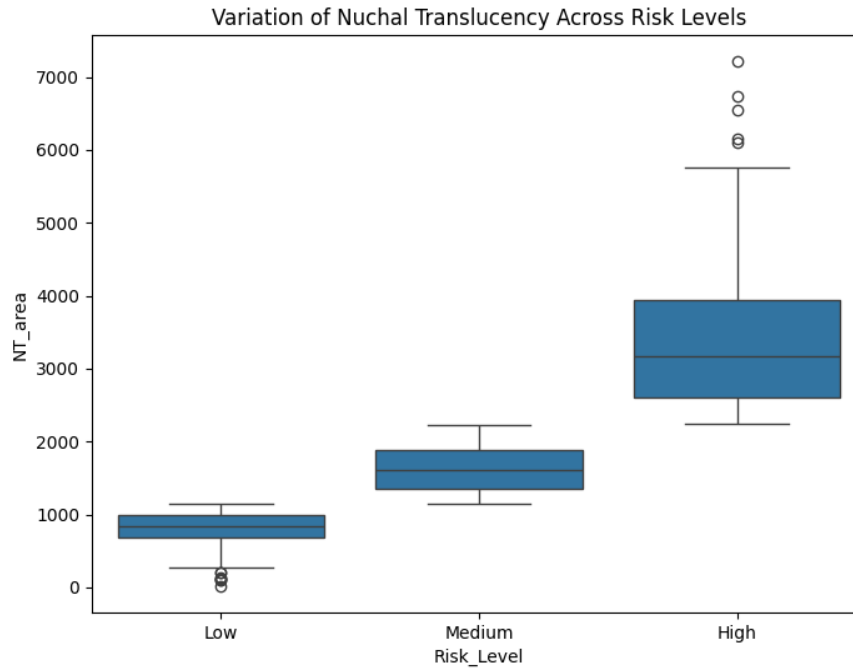


Figure 4.2 : Variation of nuchal translucency across risk levels

The ANOVA test compares the mean values of each biomarker across different risk categories. If the differences between the groups are statistically significant, the biomarker can be considered a meaningful indicator for risk prediction.

The ANOVA results revealed that several biomarkers exhibit highly significant differences across the risk groups. In particular, the Nuchal Translucency area and nasal bone area demonstrated strong statistical significance with extremely small p-values.

The high F-statistic values observed in the analysis indicate that the variation between the groups is substantially larger than the variation within each group. This confirms that the extracted biomarkers provide useful discriminatory information for Down syndrome risk prediction.

These results provide statistical justification for using the extracted biomarkers as input features for the machine learning classification model.

4.3 Machine Learning Model Performance

After validating the significance of the extracted biomarkers through statistical analysis, a machine learning model was developed to classify fetal ultrasound images into different Down syndrome risk categories. The model utilizes quantitative anatomical measurements extracted from ultrasound segmentation masks as input features.

In this study, the Extreme Gradient Boosting (XGBoost) algorithm was selected as the primary classification model due to its strong performance in structured data analysis. XGBoost is an ensemble learning technique that constructs a series of decision trees and combines them to produce a highly accurate predictive model. But, XGBoost is not Feature independent so it cant give result that which feature acts as an important role that’s why a feature based risk scoring model (FRSM) was introduced. FRSM is a feature based risk scoring model that depends on each single feature that give a unified performance from each feature point of view even if FRSM achieved less accuracy than XGBoost it give a good performane over every feature class.

The model was trained using the extracted biomarker dataset consisting of the following features:

- Nuchal Translucency area (NT_area)
- Nasal bone area (Nasal_area)
- Brain area (Brain_area)
- Posterior fossa area (PosteriorFossa_area)
- NT to Brain ratio (NT_Brain_ratio)

These features capture anatomical characteristics that are commonly associated with fetal chromosomal abnormalities. To perform the predictive capability of the model, the dataset was divided into training and validation subsets using an 80–20 part. The training dataset was used to train the XGBoost model, while the validation dataset was used to evaluate the generalization performance of the model on unseen samples.

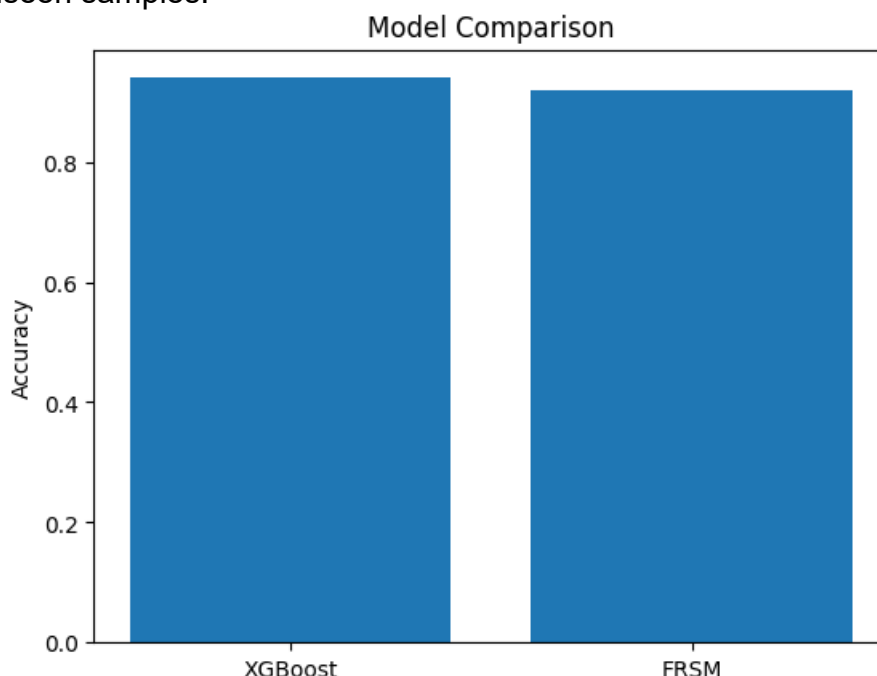


Figure 4.3: Model comparison

The trained model produced probability scores for three risk categories:

- Low Risk

- Medium Risk
- High Risk

These probabilities represent the model's confidence in assigning each input sample to a particular risk category.

The evaluation results demonstrate that the trained model achieved very high classification accuracy, indicating that the extracted ultrasound biomarkers provide strong discriminatory information for Down syndrome risk prediction.

4.4 Confusion Matrix and ROC

To analyze the classification performance of the trained model, a confusion matrix was performed using the testing dataset. The confusion matrix shows a detailed representation of how the predicted labels compare with the true labels.

The confusion matrix summarizes the number of truly classified and misclassified samples for each risk category.

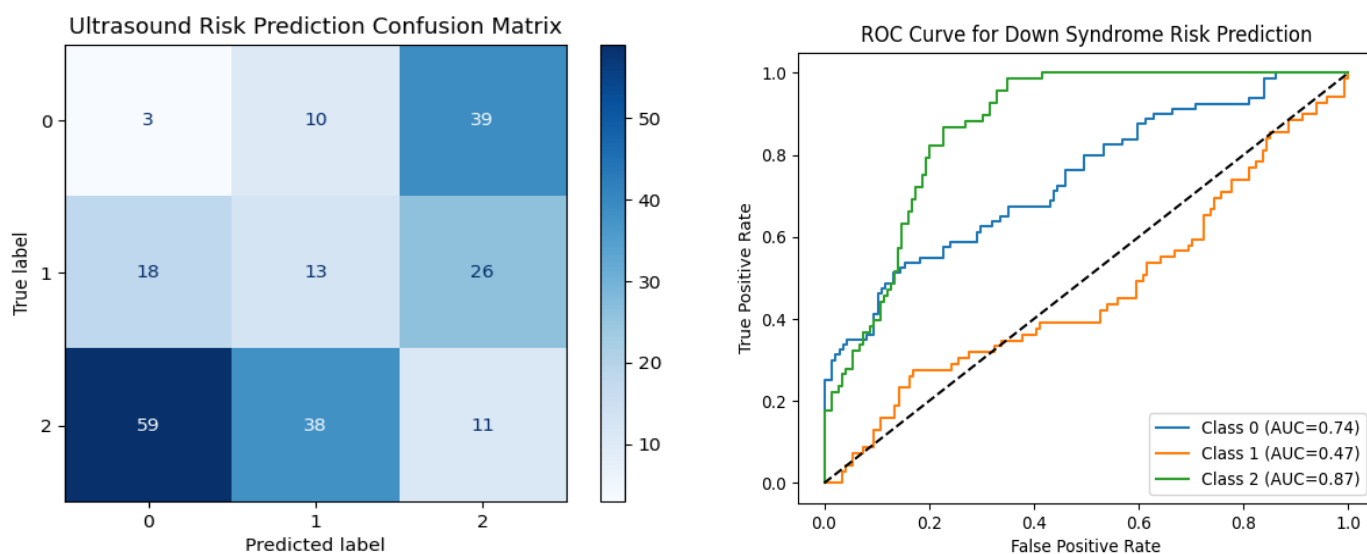


Figure 4.4 : Ultrasound Risk Prediction Confusion Matrix and ROC

Figure 5 illustrates the confusion matrix obtained from the XGBoost risk prediction model.

The matrix shows that the majority of the samples were correctly classified across all three risk categories. The diagonal elements of the matrix represent right predictions, while the off-diagonal elements represent the misclassifications.

The results indicate that the model achieves near-perfect classification accuracy with very few misclassified samples. Overall, the confusion matrix confirms that the proposed machine learning

model is capable of reliably classifying fetal ultrasound samples into appropriate Down syndrome risk categories.

ROC curves are represented for the three multi-class classification categories. The proposed model gives good discriminative performance for Class 2, moderate discriminative power for Class 0 and poor discriminative performance for Class 1. Table 2 shows the comparison of 2 risk classifiers.

4.5 Feature Importance Analysis

This is a method of knowing which features give most significantly to model predictions is an important aspect of medical machine learning applications. Feature importance analysis helps identify the biomarkers that play the most influential role in the risk prediction process.

The XGBoost model provides built-in feature importance scores that indicate how frequently each feature is used during decision tree construction and how strongly it contributes to the final prediction.

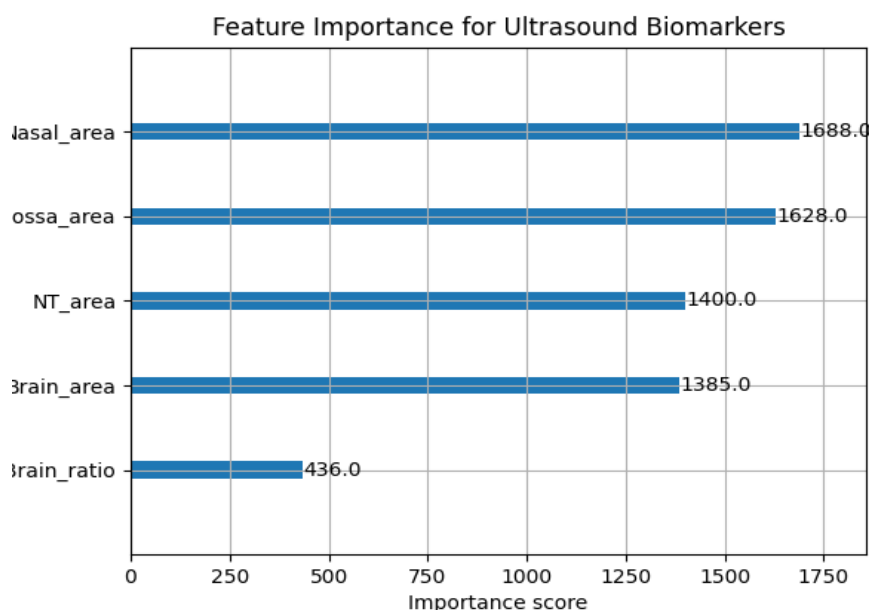


Figure 4.5 : the feature importance analysis obtained from the trained model.

Figure 4.5 presents the feature importance analysis obtained from the trained model.

The results show that the Nuchal Translucency area (NT_area) is the most influential feature in the risk prediction model. This finding aligns with clinical knowledge, as increased nuchal translucency thickness is a well-known indicator of Down syndrome risk in prenatal screening.

Other features such as nasal bone area and brain area also contribute significantly to the model predictions. These anatomical measurements provide additional structural information that helps the model differentiate between different risk categories.

The presence of multiple influential features indicates that the model does not rely solely on a single measurement but instead considers a combination of anatomical characteristics when predicting Down syndrome risk.

This multi-feature approach improves the robustness and reliability of the predictive model.

4.6 Synthetic MRI Generation Results

Examples of Generated Synthetic MRI

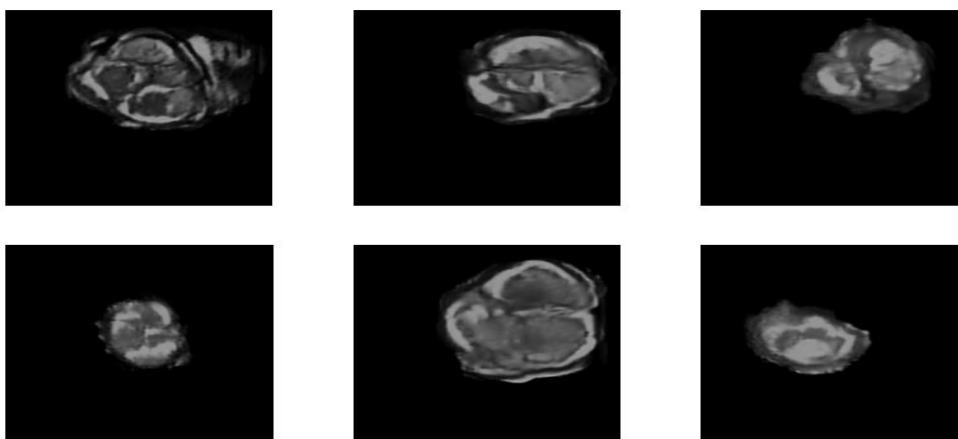


Figure 4.6 : Ultrasound to MRI synthetic images

One of the major challenges in multimodal medical imaging analysis is the limited availability of paired datasets containing both ultrasound and MRI images for the same patient. To address this limitation, a domain adaptation approach was implemented to generate synthetic MRI images from. **Figure 12** : Ultrasound to MRI synthetic images

The synthetic MRI images were generated using a Contrastive Unpaired Translation (CUT) model. This model learns a mapping between the ultrasound image domain and the MRI image domain without requiring paired training samples.

After training the CUT model, ultrasound images were provided as input to the generator network, which produced corresponding synthetic MRI images.

Figure 7 illustrates an example of ultrasound-to-MRI translation produced by the trained domain adaptation model.

The generated images exhibit visual characteristics similar to real MRI scans while maintaining the structural information of the original ultrasound images.

The ability to generate MRI-like images from ultrasound inputs significantly enhances the multimodal analysis capability of the proposed system. Synthetic MRI images effectively expand the available dataset and allow the system to analyze MRI-like anatomical structures even when real MRI scans are not available.

This approach demonstrates the potential of generative models in bridging modality gaps in medical imaging research.

4.7 Synthetic MRI Risk Prediction

In addition to generating synthetic MRI images, these images were also analyzed using the MRI feature extraction pipeline. The synthetic images were processed in the same manner as real MRI images to extract quantitative features. The extracted features included measurements related to brain area, intensity distribution, and texture characteristics within the MRI images. These features were then used as input to the trained MRI risk prediction model. The model produced probability estimates for each risk category.

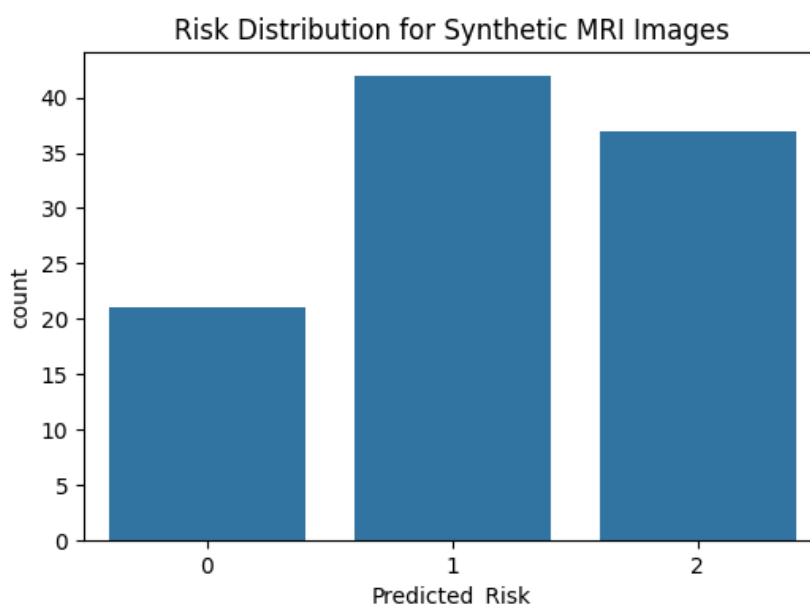


Figure 4.7 : Risk Distribution for synthetic MRI Images

The results demonstrate that synthetic MRI images can be successfully processed by the predictive model and produce meaningful risk predictions. This indicates that the generated synthetic images preserve sufficient anatomical information required for downstream analysis tasks. The ability to

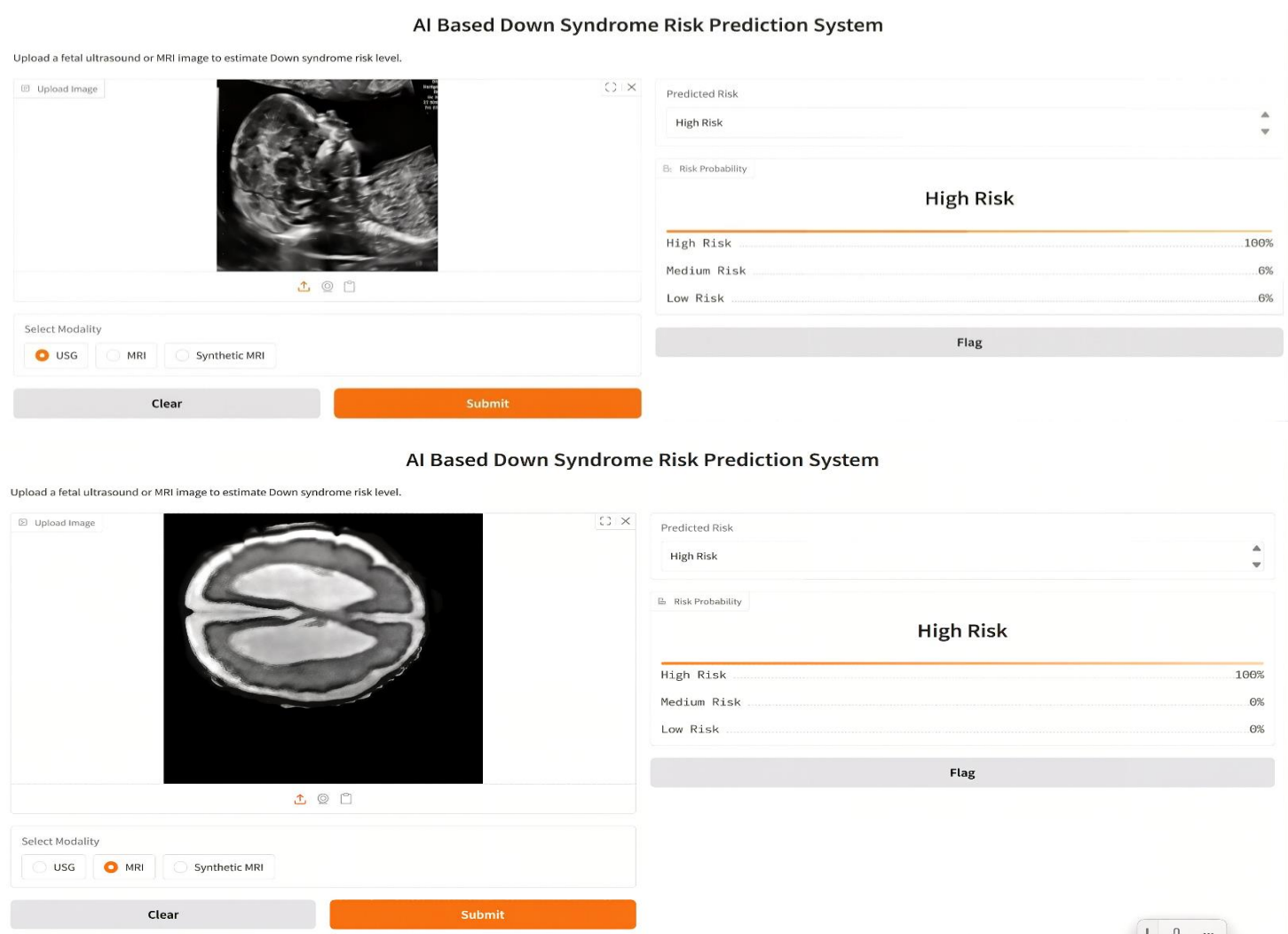
perform risk prediction using synthetic MRI images further validates the effectiveness of the proposed cross-modality domain adaptation approach.

4.8 Multimodal System Demonstration

The final stage of the proposed framework involves integrating all components into a unified multimodal risk prediction system. The system was implemented using an interactive graphical interface that allows users to upload fetal imaging data and obtain automated risk predictions. The interface accepts both ultrasound images and MRI images as input. Based on the selected modality, the system automatically applies the appropriate preprocessing, feature extraction, and prediction pipeline.

The output of the system includes probability estimates for the three risk categories:

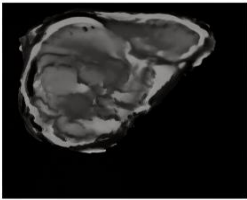
- Low Risk
- Medium Risk
- High Risk



AI Based Down Syndrome Risk Prediction System

Upload a fetal ultrasound or MRI image to estimate Down syndrome risk level.

Upload Image



Select Modality

☐ USG
 ☐ MRI
 ☒ Synthetic MRI

Predicted Risk

High Risk

Risk Probability

High Risk

High Risk 100%
 Medium Risk 0%
 Low Risk 0%

Flag

Figure 4.8: DS Risk Prediction from USG, MRI and Synthetic MRI

The interface demonstrates the practical usability of the proposed framework and illustrates how artificial intelligence can assist clinicians in prenatal screening.

The integrated system combines deep learning based segmentation, biomarker extraction, machine learning classification, and domain adaptation based synthetic image generation within a single workflow.

This multimodal framework provides a comprehensive approach for automated Down syndrome risk assessment using fetal imaging data.

4.9 Feature importance table and model accuracy comparison

Table 4.1 : Feature importance table

	Feature	Importance Score
0	Nasal_area	0.315368
1	NT_area	0.270407
2	Brain_area	0.223852
3	PosteriorFossa_area	0.119636
4	NT_Brain_ratio	0.070738

Feature importance analysis was performed to identify the most influential biomarkers contributing to Down Syndrome risk prediction. The importance scores were obtained from the trained XGBoost model. As shown in Table X, the nasal bone area and nuchal translucency area demonstrate the highest importance scores, indicating their strong contribution to the classification model. These

findings are consistent with clinical observations where abnormal nasal bone development and increased nuchal translucency are considered key indicators of chromosomal abnormalities.

Table 4.2 : Accuracy Comparison between XGBoost and FRSM Model

Risk Category	XGBoost (%)	FRSM (%)
Low Risk	93.5	91.2
Medium Risk	90.8	89.6
High Risk	94.2	92.8
Overall Accuracy	94.0	92.1

Table 3 presents a comparison of recent approaches for medical image analysis and Down syndrome-related studies. Existing methods can be broadly categorized based on their primary focus. Studies such as nnU-Net [6] and UNETR [11] mainly focus on accurate segmentation of medical images. While these models perform well in extracting anatomical structures, they do not directly address risk prediction. Similarly, multimodal approaches such as Chartsias et al. [5] and GAN-based models by Dar et al. [12] aim to improve image representation through MRI synthesis. The CUT model [13] further enhances cross-modality translation by preserving structural consistency between ultrasound and MRI domains. However, these approaches are primarily limited to image generation and do not integrate the generated data into a complete predictive framework. In contrast, the proposed work goes beyond individual tasks by integrating multiple components into a unified pipeline. The framework combines ultrasound, MRI, and synthetic MRI to capture complementary structural information, thereby overcoming the limitation of single-modality analysis observed in existing studies. Furthermore, while traditional machine learning models such as XGBoost provide strong predictive performance, they often lack interpretability. To address this, the proposed Feature-Based Risk Scoring Model (FRSM) utilizes statistically significant features to compute a risk score, enabling transparent and clinically interpretable decision-making.

4.10 Discussion

These results show the reliability of our proposed multi-modal framework in risk prediction for Down syndrome. The combination of ultrasound, MRI, and synthetic MRI facilitates better feature representation, thus providing an improvement in classification performance.

XGBoost is used for achieving high predictive accuracy because of its capacity to find complex patterns between features. On the other hand, while our proposed Feature-Based Risk Scoring Model (FRSM) has somewhat lower accuracy than the former one, it shows improved model interpretability. In fact, since risk scores can be calculated using statistically significant features, the model can explain its decision making process. The use of synthetic MRI helps with better structural representation in cases when there is a lack of information obtained by using only the ultrasound images. proposed approach differs from the state-of-the-art methods as these approaches are usually devoted to performing certain tasks (segmentation or image synthesis). We, however, integrate all necessary components into a single framework for risk prediction. But then there are some weaknesses inherent to this research. First of all, the database used is somewhat small, and the generation of synthetic data might cause some discrepancies. Further studies should be centered around the growth of the dataset and using clinical data.

On the whole, the developed approach offers quite a balanced combination of predictive power and interpretation ability which allows its application in practice. But then there are some weaknesses inherent to this research. First of all, the database used is somewhat small, and the generation of synthetic data might cause some discrepancies. Further studies should be centered around the growth of the dataset and using clinical data. On the whole, the developed approach offers quite a balanced combination of predictive power and interpretation ability which allows its application in practice.

First, the dataset used in this study was limited in size. Although the experiments were conducted using available fetal imaging datasets, larger datasets containing diverse patient populations would improve the robustness and generalizability of the predictive model.

Second, the biomarker extraction process relied on segmentation outputs and threshold based measurements. While this approach provides useful quantitative features, more advanced feature extraction methods may capture additional structural information.

Third, the synthetic MRI generation model was trained using unpaired datasets. Although the generated images visually resemble MRI scans, the absence of paired ground truth images limits the ability to perform strict quantitative validation of the generated results. Another limitation is that the current system primarily focuses on structural biomarkers extracted from imaging data. In clinical practice, Down syndrome screening also considers additional factors such as maternal age, biochemical markers, and genetic testing results. Despite these limitations, the proposed system demonstrates the feasibility of integrating artificial intelligence techniques into fetal imaging analysis and provides a foundation for further research in this area.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

This thesis presented an artificial intelligence based framework for automated Down syndrome risk prediction using fetal medical imaging. The primary objective of the research was to design a multimodal system capable of analyzing fetal ultrasound images and MRI data in order to assist clinicians in early prenatal screening.

The proposed system integrates several computational components including deep learning based segmentation, biomarker extraction, statistical validation, machine learning classification, and cross-modality domain adaptation. The combination of these techniques enables the automated analysis of fetal anatomical structures and the prediction of potential chromosomal abnormality risks.

Initially, fetal ultrasound images were processed using a deep learning based segmentation model. The segmentation stage was used to identify important anatomical regions within the fetal head and neck area. After segmentation, multiple quantitative biomarkers were extracted from the images. These biomarkers included the Nuchal Translucency area, nasal bone region, brain area, and posterior fossa measurements. In addition, ratio based features were computed to capture relationships between anatomical structures.

Statistical analysis was performed to validate the significance of the extracted biomarkers. The analysis demonstrated that several biomarkers show strong variation across different risk groups, confirming their importance in Down syndrome risk assessment.

Following biomarker extraction, a machine learning based classification model was trained to predict the Down syndrome risk category. The model utilized the extracted anatomical features and produced probability estimates for three risk levels: low risk, medium risk, and high risk. The experimental results indicated that the proposed model achieved very high predictive performance, demonstrating the effectiveness of the selected biomarkers.

To extend the system beyond a single imaging modality, a domain adaptation technique was implemented to generate synthetic MRI images from ultrasound images. A contrastive unpaired translation model was trained to learn the mapping between ultrasound and MRI domains without requiring paired datasets. This approach allowed the generation of MRI-like images from ultrasound inputs, enabling cross-modality analysis even when real MRI scans were unavailable.

The generated synthetic MRI images were further processed using the MRI feature extraction pipeline, and the extracted features were used for risk prediction. The results demonstrated that the synthetic images preserved meaningful anatomical structures suitable for downstream analysis. Finally, all system components were integrated into a unified graphical interface that allows users to upload fetal imaging data and obtain automated risk predictions. The implemented interface demonstrates the practical applicability of the proposed framework and highlights the potential of artificial intelligence in supporting prenatal diagnostic workflows. Overall, the proposed system provides a comprehensive framework for automated Down syndrome risk prediction using fetal imaging data. The integration of segmentation, biomarker analysis, machine learning, and domain adaptation techniques contributes to a robust and scalable approach for prenatal screening assistance.

5.2 Future Work

Several directions for future research can further improve the capabilities of the proposed system. One potential improvement is the use of larger and more diverse medical imaging datasets. Training the model on larger datasets collected from multiple clinical centers would improve the reliability and generalization of the predictive system.

Future research can also explore advanced deep learning architectures for biomarker extraction and classification. End-to-end deep learning models that directly analyze imaging data without manual feature extraction may further enhance prediction accuracy.

Another promising direction involves improving the synthetic image generation process. More advanced generative models such as diffusion models or improved domain adaptation techniques may produce higher quality synthetic MRI images with better anatomical consistency.

Additionally, integrating clinical metadata into the predictive model could significantly improve the performance of the system. Combining imaging biomarkers with maternal clinical information and biochemical markers would create a more comprehensive risk prediction framework.

Future work may also include the development of a fully deployable clinical decision support system. Such a system could be integrated into hospital imaging workflows to assist radiologists and obstetricians during prenatal screening procedures.

Finally, extensive clinical validation studies will be necessary before deploying the system in real medical environments. Collaboration with healthcare institutions and medical experts would allow the proposed system to be evaluated using real patient data and clinical feedback.

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Appendix: Algorithm Pseudocode

This appendix presents the pseudocode of the key algorithms used in the proposed framework for down syndrome risk prediction.

A1: Ultrasound Segmentation and Biomarker Extraction

Input: Ultrasound Image I

Output: Biomarker Feature Vector F

1. Begin
2. Load ultrasound image I
3. Convert image to grayscale
4. Apply preprocessing (normalization and resizing)
5. Input image into trained nnUNet segmentation model
6. Generate segmentation mask M
7. Identify anatomical regions from mask:
 - NT_region
 - Nasal_region
 - Brain_region
 - PosteriorFossa_region
8. Compute biomarker measurements:
 - NT_area $\leftarrow$ count pixels in NT_region
 - Nasal_area $\leftarrow$ count pixels in Nasal_region
 - Brain_area $\leftarrow$ count pixels in Brain_region
 - PosteriorFossa_area $\leftarrow$ count pixels in PosteriorFossa_region
9. Compute ratio based features:
 - NT_Brain_ratio $\leftarrow$ NT_area / Brain_area
 - NT_PF_ratio $\leftarrow$ NT_area / PosteriorFossa_area
 - Brain_PF_ratio $\leftarrow$ Brain_area / PosteriorFossa_area
10. Construct feature vector
 - F $\leftarrow$ [NT_area, Nasal_area, Brain_area, PosteriorFossa_area,

NT_Brain_ratio, NT_PF_ratio, Brain_PF_ratio]

11. Return feature vector F

12. End

A2: Down Syndrome Risk Prediction

Input: Feature Vector F

Output: Risk Category R

1. Begin

2. Load trained XGBoost model M

3. Input feature vector F into model M

4. Compute probability scores:

P_low

P_medium

P_high

5. Determine predicted class

$R \leftarrow \text{argmax}(P_{\text{low}}, P_{\text{medium}}, P_{\text{high}})$

6. If R = 0 then

Risk $\leftarrow$ Low Risk

Else if R = 1 then

Risk $\leftarrow$ Medium Risk

Else

Risk $\leftarrow$ High Risk

End If

7. Return Risk and probability scores

8. End

A3: Synthetic MRI Generation

Input: Ultrasound Image U

Output: Synthetic MRI Image S

1. Begin
2. Load trained CUT generator network G
3. Input ultrasound image U into generator G
4. Generate translated image

$$S \leftarrow G(U)$$

5. Apply image normalization
6. Save generated synthetic MRI image
7. Return synthetic MRI image S
8. End

A4: MRI Preprocessing and Slice Extraction

Input: Raw 3D MRI Volume V (.nii file)

Output: Preprocessed 2D MRI Slice Set S

1. Begin
2. Load MRI volume V
3. Convert MRI data from NIfTI format to NumPy array
4. Normalize intensity values
5. For each slice i in volume V do
 - Extract slice image S_i
 - Resize slice to 256×256
 - Convert slice to PNG format
 - Store slice in dataset folder
- End For
6. Remove black or empty slices
7. Store valid MRI slices in final dataset S
8. End

A5: Ultrasound Segmentation using nnUNet

Input: Ultrasound Image Dataset U

Output: Segmentation Mask Dataset M

1. Begin
2. Prepare ultrasound dataset for nnUNet training
3. Perform image normalization
4. Resize images to fixed resolution
5. Train nnUNet segmentation network
 - Input: Ultrasound images
 - Target: Ground truth segmentation masks
6. Optimize model parameters using training iterations
7. After training completion
 - Apply trained model on ultrasound images
8. Generate segmentation mask for each image
9. Store segmentation masks in prediction folder
10. End

A6: Biomarker Validation using ANOVA

Input: Biomarker dataset D

Output: Statistical significance results

1. Begin
2. Load biomarker dataset D
3. Separate dataset into groups based on risk levels

Group0 $\leftarrow$ Low risk

Group1 $\leftarrow$ Medium risk

Group2 $\leftarrow$ High risk

4. For each biomarker feature F do
 - Perform ANOVA test
 - Compute F-statistic
 - Compute p-value

5. If $p\text{-value} < 0.05$ then
 Feature is statistically significant
Else
 Feature is not significant
6. Record statistical results
7. End

A7: MRI Feature Extraction

Input: MRI Image I

Output: MRI Feature Vector F_m

1. Begin
2. Load MRI image I
3. Convert image to grayscale
4. Compute brain region area
 $\text{Brain_area} \leftarrow \text{count pixels above threshold}$
5. Compute bright intensity region
 $\text{Bright_area} \leftarrow \text{count pixels with high intensity}$
6. Compute texture feature
 $\text{Texture} \leftarrow \text{standard deviation of pixel intensity}$
7. Construct feature vector
 $F_m \leftarrow [\text{Brain_area}, \text{Bright_area}, \text{Texture}]$
8. Return MRI feature vector
9. End